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The Sudden Implosion of Silicon Valley Bank

(SVB) failed because of a textbook case of mismanagement by the bank. [1]

—Michael S. Barr, vice chair for supervision, US Federal Reserve

This past week we saw the biggest bank failure in more than 15 years as federal regulators seized Silicon Valley Bank. This is a classic asset-liability mismatch.[2]

—Larry Fink, cofounder, chair, and CEO, BlackRock, Inc.

On March 12, 2023, regulators[3] closed Silicon Valley Bank (SVB) after what could be described as an unprecedented, if not portentous, social-media-driven bank run.[4] "The outflows at SVB were without equal," described one economist.[5] Indeed, depositors withdrew $42 billion in less than 24 hours, forcing regulator action and the bank's closure.[6] News reports characterized the collapse as a shockwave event unnerving the banking and technology industries.

Although not a household name, SVB was among the top 20 largest banks in the United States.[7] SVB catered to start-up and early-stage companies that maintained banking relationships with SVB as they grew and scaled their businesses. It banked nearly 50% of all US venture-backed technology and life science companies.[8]

[1] Michael S. Barr, "Re: Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank," Board of Governors of the Federal Reserve System, April 28, 2023, https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf (accessed June 30, 2023).

[2] "Larry Fink's Annual Chairman's Letter to Investors," BlackRock, Inc., https://www.blackrock.com/corporate/investor-relations/larry-fink-annual-chairmans-letter# (accessed July 7, 2023).

[3] The California Department of Financial Protection and Innovation shut down the bank and appointed the Federal Deposit Insurance Corporation as the receiver.

[4] Hannah Lang, "Analysis: Social Media-Driven Bank Runs Burden Regulators with a Bigger Problem," Reuters, March 22, 2023, https://www.reuters.com/business/finance/social-media-driven-bank-runs-burden-regulators-with-bigger-problem-2023-03-21/ (accessed June 11, 2023).

[5] https://www.reuters.com/business/finance/social-media-driven-bank-runs-burden-regulators-with-bigger-problem-2023-03-21/.

[6] https://www.reuters.com/business/finance/social-media-driven-bank-runs-burden-regulators-with-bigger-problem-2023-03-21/.

[7] Jaclyn Diaz, "For 40 Years, Silicon Valley Bank Was a Tech Industry Icon. It Collapsed in Just Days," NPR, March 15, 2023, https://www.npr.org/2023/03/15/1163269781/silicon-valley-bank-svb-collapse-history (accessed June 6, 2023).

[8] "SVB Q4 2022 Financial Highlights," January 19, 2023, https://s201.q4cdn.com/589201576/files/doc_financials/2022/q4/Q4_2022_IR_Presentation_vFINAL.pdf (accessed June 12, 2023).

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Moreover, SVB also custodied assets for some 1,074 private equity (PE) and venture capital (VC) funds in recent years.[9]

The fall of the iconic SVB[10] was the largest US bank failure since the 2008 global financial crisis, and the first after the implementation of increased oversight and supervision of financial institutions deemed by the US government to be systemically important or "too big to fail."[11] In 2018, however, a bipartisan rollback of some bank regulations on smaller-sized banks, meaning those with assets of $250 billion or less, perhaps played a role in SVB's troubles. Following SVB's collapse, several other smaller-sized US banks also failed. Clients, industry leaders in finance and technology, legislators, and regulators all had the same question: What happened?

A Different Kind of Bank

SVB opened for business in 1983 as the principal subsidiary of Silicon Valley Bancshares, which had been incorporated a year earlier. The founders' intent was to operate a bank that would be a lender to early-stage, technology-oriented companies that had yet to show a profit but did have growth potential.[12] At the time, Silicon Valley, in Northern California, south of San Francisco, was an embryo of what it would become: one of the world's epicenters for high-tech start-ups, entrepreneurs, and innovators. As SVB's first CEO Roger Smith explained, "Silicon Valley was not [well] known in 1983."[13]

SVB's approach to lending to start-ups (known as venture debt) defied conventional bank-lending practices, in which companies (borrowers) needed to demonstrate assets and profitability to be considered creditworthy.[14] Wrote one reporter,

> Historically, traditional banks have had a hard time trying to figure out what to do with tech startups, particularly ones founded by a couple of young people with little credit history, no revenue, and only a few hundred thousand dollars in funding from a few rich people to their name…Oftentimes, such startups would get turned away or offered nothing more than a basic checking account…Whereas more traditional bankers wanted to see balance sheets and income statements, SVB would consider more atypical measures, like the founder experience, burn rate, business plan, runway, and cap table.[15] Crucially, they also carefully considered who the venture capitalists were that were backing the startup…SVB's willingness to provide young startups with lines of credit also proved beneficial, allowing startups to push to grow faster and harder and make the most of their funding.[16]

Smith elaborated,

[9] Alexandra Sternlicht and Jessica Mathews, "An Analysis of Regulatory Filings Reveals That 1,074 Firms—from Andreessen Horowitz to General Catalyst—Were Holding Capital at Silicon Valley Bank in 2022," *Fortune*, March 12, 2023, https://fortune.com/2023/03/12/venture-firms-custodied-silicon-valley-bank/ (accessed July 7, 2023).

[10] SVB, a California state-chartered bank, was the principal subsidiary of Silicon Valley Bank Financial Group (SVBFG), accounting for around $209 billion of the SVBFG's total assets of approximately $212 billion.

[11] https://www.npr.org/2023/03/15/1163269781/silicon-valley-bank-svb-collapse-history.

[12] Robert "Bob" Medearis and Roger Smith, interview by Stephen "Steve" Smith, "Silicon Valley Bank Oral History Panel," Computer History Museum, November 11, 2014, https://archive.computerhistory.org/resources/access/text/2015/07/102739977-05-01-acc.pdf (accessed June 7, 2023).

[13] https://archive.computerhistory.org/resources/access/text/2015/07/102739977-05-01-acc.pdf.

[14] "Silicon Valley Bank Celebrates 20 Years of Dedication to Entrepreneurs," SVB, October 17, 2023, https://www.svb.com/news/company-news/silicon-valley-bank-celebrates-20-years-of-dedication-to-entrepreneurs (accessed June 6, 2023).

[15] A cap table (capitalization table) was a spreadsheet listing the securities (i.e., common shares, preferred shares, warrants) in a start-up or early-stage company. It identified ownership percentages, prices paid by investors, and the value of the securities.

[16] Maxwell Strachan, "How Silicon Valley Bank Made a Business out of Making VCs Happy," Vice, March 16, 2023, https://www.vice.com/en/article/pkayyg/how-silicon-valley-bank-made-a-business-out-of-making-vcs-happy (accessed June 7, 2023).

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> [W]e were not venture capitalists, and we made that very clear, that our money—we loaned the money,[17] and we want it back, right? And so we did it smartly, mainly in the early days, on receivables. And so here, you're a new company, but you have a receivable from HP. That's a pretty good piece of paper…a lot of people worried about us because they thought we were taking too high a risk, where in fact we were not.[18]

While SVB had used receivables in the very early days, the firm came to rely on a client company's access to VC—the ability to raise additional equity—as the primary repayment source for the loan, not historical cash flow or working capital assets. VC was generally used to support start-ups and other businesses with substantial growth potential through the early and middle stages of development. Many funds focused on specific industries or sectors, geographies, or stages of company development.[19] VC funds, meaning private funds managed by a VC firm, raised money from investors, typically institutional investors (e.g., pension funds, endowments, family offices, etc.), known as limited partners (LPs). VC funds were usually structured to last at least 10 years. In exchange for money and mentorship, a fund received an equity stake in the business, which was generally locked in until a liquidity event (e.g., acquisition or IPO) when the VC firm and its fund investors hoped to realize profits from their investment. Venture debt came after equity and venture lenders, such as SVB, used VC support as "a source of validation and the primary yardstick for underwriting a loan," explained an SVB vice president. [20]

In 1986, SVB merged with National Intercity Bancorp and completed an IPO in 1988.[21] SVB's first decade primarily focused on commercial lending to start-up and early-stage companies within a variety of technology and life sciences industries, producing semiconductors, electronics, software, communications, peripherals, and medical devices. In the mid-1990s, the bank expanded offices across parts of California and to other states (Maryland, Massachusetts, and Oregon) and identified four underserved niche markets to which it could provide loans: commercial real estate, entertainment (e.g., loans to make movies), religious institutions, and premium wineries. In 2002, SVB exited all niches except for premium wineries.

In 2005, Silicon Valley Bancshares was rebranded as SVB Financial Group (SVBFG), as the firm had expanded beyond commercial banking. By the turn of the new millennium, while SVB had continued to serve start-ups and emerging companies in the technology and life science industries, along with premium wineries, SVBFG had expanded into investment banking (M&A services), merchant banking (PE and VC fund investments), and private banking for high-net-worth individuals (providing loans, personal asset management, trust and estate planning, and mortgages).

A key component of SVB's business strategy was to develop relationships with client companies at an early stage and to offer them banking products and services throughout their business life cycle. This included cultivating strong relationships with VC firms, many of which were also clients, thereby providing access to many other potential clients (i.e., other companies in their portfolios).[22] Indeed, over time, SVB became a banking partner not only to start-up companies but also to their founders, and to VCs and their funders. "[SVB] just tied it all together in this really comprehensive offering that serviced the startups, the founders for their personal needs, the VCs for their funds, the VCs for their personal lives, so they can have money to invest in

[17] Initially, loans were funded by investors in the bank.
[18] https://archive.computerhistory.org/resources/access/text/2015/07/102739977-05-01-acc.pdf.
[19] Vanessa Kuhlor, "What Is Venture Capital?," SVB, https://www.svb.com/startup-insights/vc-relations/what-is-venture-capital (accessed June 12, 2023).
[20] Armando A. Argueta, "Understanding Venture Debt Financing," Silicon Valley Bank, https://www.svb.com/startup-insights/venture-debt/how-does-venture-debt-work (accessed June 28, 2023).
[21] https://www.svb.com/news/company-news/silicon-valley-bank-celebrates-20-years-of-dedication-to-entrepreneurs.
[22] SVB Financial Group SEC 10-K Report, 2000.

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their funds," said one San Francisco–based VC.[23] Examples included providing credit cards and home mortgages. SVB was known for its "white glove" treatment of its largest depositors.

One observer wrote,

> SVB also leaned heavily into the growing venture capital investors who backed startups. By partnering with venture capitalists who would back their startups on their successive funding rounds the bank reduced their deposit and venture debt risk. And SVB did this and had loan losses comparable to traditional banks, yet without traditional collateral requirements of major banks. This "magic triangle" of startups, VCs, and SVB created a symbiotic relationship and worked beautifully for 40 years, helping launch tech bellwethers like Cisco, Etsy, and Roku.[24]

SVB opened its first international location in Israel in 2008, and in the ensuing years it added offices in the United Kingdom, Ireland, Germany, Canada, and Denmark.

Despite SVBFG's expansion into other financial services, SVB continued to be the group's largest operating segment, which in 2023 also included SVB Private, SVB's private bank and wealth-management division. Other operating segments were SVB Capital and SVB Securities. SVB Capital was SVBFG's in-house VC arm that invested in venture funds and also made direct investments in equity rounds sponsored by other VC firms. SVB Securities, the group's wholly owned investment bank, provided capital-raising services, M&A advising, and structured finance products and services to SVB's clients; it also did equity research (coverage of over 360 health care and technology companies), sales, and trading. SVBFG's acquisitions included health care and life science investment bank Leerink Partners, which it acquired for $280 million in upfront cash in 2019,[25] and Boston Private, a provider of wealth-management, trust, and banking services, which it acquired for approximately $900 million in 2021 and which became part of SVB Private.[26] (**Exhibits 1**, **2**, and **3** show SVBFG's operational and financial data from 2020 to 2022; **Exhibit 4** shows SVB's loan portfolio; and **Exhibit 5** shows peer comparison on several industry metrics.) Over the four-year period from January 2018 to January 2022, SVBFG's share price had outperformed benchmarks, before faltering in the remainder of 2022 and into 2023. (See **Exhibit 6**.)

Culture

According to some SVB employees, the firm's culture was more like a tech company's than a bank's. One SVB associate wrote,

> [F]ree lunches and cool hiring initiatives were motivating, but sometimes it felt like the company was being run more like a startup than a bank. But at the same time, I appreciated that unlike other finance institutions, our culture wasn't super cutthroat, and you didn't see many finance/tech "bro" personalities running around. Everyone seemed to have values of trying to build a good business for a better world.[27]

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[23] https://www.vice.com/en/article/pkayyg/how-silicon-valley-bank-made-a-business-out-of-making-vcs-happy.

[24] Lou Shipley, "What Silicon Valley Bank Did Right," *Harvard Business Review*, March 24, 2023, https://hbr.org/2023/03/what-silicon-valley-bank-did-right (accessed June 11, 2023).

[25] "SVB Financial Group Completes Acquisition of Healthcare Investment Bank Leerink Partners," SVB Financial Group press release, January 4, 2019, https://ir.svb.com/news-and-research/news/news-details/2019/SVB-Financial-Group-Completes-Acquisition-of-Healthcare-Investment-Bank-Leerink-Partners/default.aspx (accessed June 13, 2023).

[26] "SVB Financial Group to Acquire Boston Private, a Leading Provider of Wealth Management, Trust and Banking Services," SVB Financial Group press release, January 4, 2021, https://www.svb.com/news/company-news/svb-financial-group-to-acquire-boston-private-a-leading-provider-of-wealth-management-trust-and-banking-services2 (accessed June 13, 2023).

[27] Emmalyse Brownstein, "I'm an Associate at SVB. I Love the Culture of This Company, but I'm Livid at the Failures of Management.," *Business Insider*, March 23, 2023, https://www.businessinsider.com/svb-employee-explains-disappointment-anger-with-the-company-2023-3 (accessed July 10, 2023).

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Others noted a focus on social justice issues, with weekly internal "TED talks" on social issues and classes on "how to make sure you were not committing a microaggression."[28] And while many financial institutions called its workforce back to the office, SVB continued to embrace remote working after the COVID-19 pandemic. In fact, its senior team worked from locations from Hawaii to New York. "It is harder to have a challenging call over Zoom," said one academic who studied remote working. "It makes it harder to challenge management. Ideas like hedging interest rate risk often come up over lunch or in small meetings."[29]

In 2021, SVB acquired a financial technology (fintech) start-up called Standard Treasury to help accelerate the development and expansion of its digital banking platform. "When we sold Standard Treasury to SVB, we found ourselves up against a culture that sought consensus at the expense of decision-making," said one of Standard Treasury's cofounders.[30]

Setting the Stage: 2020 to 2022

Venture capital

In 2022, US-based VC funds raised a record amount—$163 billion—just a year after raising $155 billion in 2021, which was a 75% increase over 2020. [31] VC investments (known as deals) in companies dropped from a high of $337 billion in 2021 to $212 billion in 2022—still a historically significant amount. (See **Exhibit 7**.) "While numerous events may have contributed to the drop in activity, the Fed's benchmark interest rate hikes are the most proximate cause for the decrease in VC activity," wrote the National Venture Capital Association (NVCA).[32]

Meanwhile in 2022, the IPO market for VC-backed US companies went cold. IPOs in that year raised only $1.6 billion, compared with $60 billion the year before, which was double the amount raised in 2020.[33] VC-backed US company IPOs in 2020, 2021, and 2022 numbered 109, 191, and 36, respectively.[34] That trend mirrored the market capitalization for US technology stocks in general (see **Exhibit 8**).

In Silicon Valley, VC investment in 2021 amounted to $44 billion, a record high.[35] This compared to $26.4 billion in 2020,[36] a new record at the time, and $18 billion in 2019.[37] In early 2020, the COVID-19 pandemic hit the United States, disrupting the way Americans worked and lived. The pandemic, however, was a boon to investments into Silicon Valley–based companies, as entrepreneurs either pivoted their start-ups or launched new companies to respond to shifting consumer behavior under lockdowns and other restrictions. This included investments in digital platforms and new digital technologies. The top deals in 2021 included

[28] Tabby Kinder and Antoine Gara, "'It Is Not Cut-Throat Like Goldman Sachs': SVB's Culture in Focus," *Financial Times*, March 16, 2023, https://www.ft.com/content/6e23a2fb-484e-418d-b309-bf558b3a6a17 (accessed June 30, 2023).

[29] https://www.ft.com/content/6e23a2fb-484e-418d-b309-bf558b3a6a17.

[30] Zac Townsend, "Signals: The Silicon Valley Bank That Never Was," This Week in Fintech, March 22, 2023, https://www.thisweekinfintech.com/signals-the-silicon-valley-bank-that-never-was-2/ (accessed July 10, 2023).

[31] "NVCA 2023 Yearbook," National Venture Capital Association (NVCA), https://nvca.org/wp-content/uploads/2023/03/NVCA-2023-Yearbook_FINALFINAL.pdf (accessed June 16, 2023).

[32] https://nvca.org/wp-content/uploads/2023/03/NVCA-2023-Yearbook_FINALFINAL.pdf.

[33] "Value of Venture Capital–Backed IPOs in the United States from 2000 to 2022 (in billion U.S. dollars)," (graph) Statista, citing WilmerHale, April 6, 2023, SEC Filings, 2000 to 2022, https://www-statista-com.proxy1.library.virginia.edu/statistics/251187/venture-capital-backed-ipos-in-the-united-states/ (accessed June 16, 2023).

[34] "Venture Capital in North America," Statista, https://www-statista-com.proxy1.library.virginia.edu/study/29476/venture-capital-statista-dossier/ (accessed June 16, 2023).

[35] "2022 Silicon Valley Index," Joint Venture Silicon Valley Board of Directors and Institute for Regional Studies, https://jointventure.org/images/stories/pdf/index2022.pdf (accessed June 20, 2023).

[36] "2021 Silicon Valley Index," Joint Venture Silicon Valley Board of Directors and Institute for Regional Studies, https://jointventure.org/images/stories/pdf/index2021.pdf (accessed June 20, 2023).

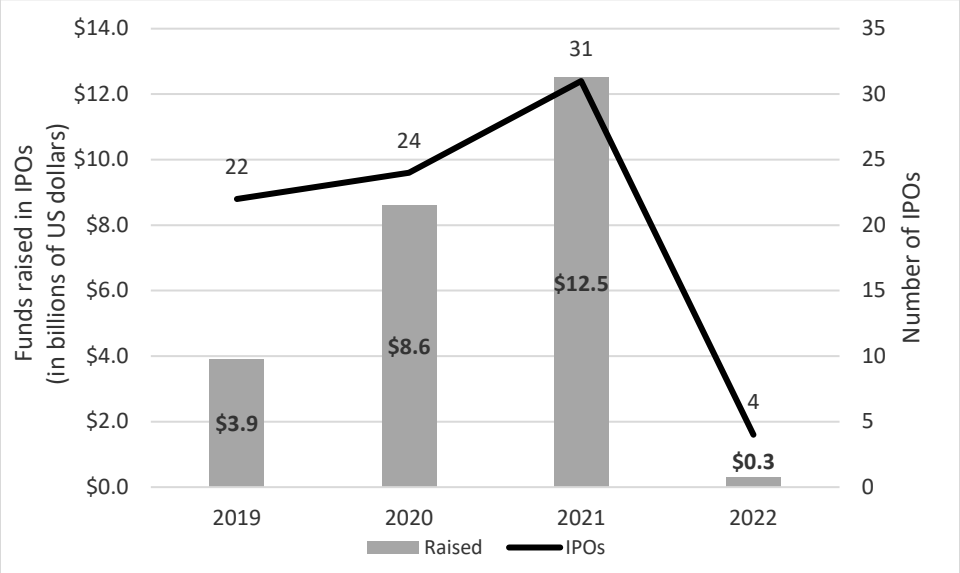
[37] "2020 Silicon Valley Index," Joint Venture Silicon Valley Board of Directors and Institute for Regional Studies, https://jointventure.org/images/stories/pdf/index2020.pdf (accessed June 20, 2023).

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investments of $3.4 billion in Robinhood, a tech-financial services firm; $2.5 billion in Waymo, an autonomous vehicle company and Google spinoff; and $1.3 billion in Lacework, which specialized in machine learning (ML) for cloud security.[38] In 2022, however, VC funding of Silicon Valley companies fell to $20.7 billion.[39] The largest deal ($3 billion) went to Altos Labs, a biotechnology cell therapy research company.[40] As far as Silicon Valley company IPOs, there was a drop-off in 2022 compared to the prior years (see **Figure 1**).

Figure 1. IPO Volumes for Silicon Valley companies (2019 to 2022).



Source: Created by authors. Data from "2023 Silicon Valley Index," Joint Venture Silicon Valley Board of Directors and Institute for Regional Studies, https://jointventure.org/images/stories/pdf/index2023.pdf (accessed June 20, 2023).

Inflation and interest rates

Beginning in early 2021, the United States began to experience inflation, a rise in prices across the economy. Experts cited reasons including low unemployment, the release of pent-up demand accrued during the COVID-19 pandemic, supply chain issues, and Russia's war on Ukraine (and its impact on energy supply). The annual inflation rate in 2022 (using the consumer price index) was 8%—the highest rate since the early 1980s.[41] To combat inflation, the US Federal Reserve (Fed) sprang into action and began to raise the federal funds rate (the interest rate banks charged one another for overnight borrowings to meet capital requirements). The Fed raised the federal funds rate at one of the fastest paces in US history[42] in its attempt to control inflation (see **Exhibit 9**). An increase to the federal funds rate led to an increase in the interest rates banks charged consumers for loans and credit cards. Higher interest rates were intended to curb consumer demand and lower inflation.

[38] https://jointventure.org/images/stories/pdf/index2022.pdf.

[39] "2023 Silicon Valley Index," Joint Venture Silicon Valley Board of Directors and Institute for Regional Studies, https://jointventure.org/images/stories/pdf/index2023.pdf (accessed June 20, 2023).

[40] https://jointventure.org/images/stories/pdf/index2023.pdf.

[41] "Consumer Price Index 1913–," Federal Reserve Bank of Minneapolis, https://www.minneapolisfed.org/about-us/monetary-policy/inflation-calculator/consumer-price-index-1913- (accessed Oct. 12, 2023).

[42] Mark Maurer, "Banks, Investors Revive Push for Changes to Securities Accounting after SVB Collapse," *Wall Street Journal*, March 20, 2023, https://www.wsj.com/articles/banks-investors-revive-push-for-changes-to-securities-accounting-after-svb-collapse-99caa9ce (accessed June 14, 2023).

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SVB deposits and investments

Between money raised via IPOs, special purpose acquisition companies (SPACs), and VC funding, particularly in 2021, SVB's client companies were flush with cash. SVB deposits increased from $62 billion on December 31, 2019, to $189 billion by the end of fiscal year 2021. Deposits fell slightly to $173 billion at the end of fiscal year 2022. Of total deposits, 30% were from early-stage tech companies and 9% were from early-stage life science or health care companies.[43]

According to one regulatory filing, more than 93% of these deposits were not insured by the government.[44] The Federal Deposit Insurance Corporation (FDIC) insured only bank deposits up to $250,000.[45] "[SVB] would seem perfectly at home in the 1920s or the 1870s: Lots of deposits and a lot of local customers," summed one scholar. "The very biggest banks have much more diversified client bases, and also somewhat more diversified funding sources. So it wouldn't be as funded as much by deposits."[46] Deposits, nonetheless, were the cheapest source of funding for banks (some deposits were not interest bearing). (See **Exhibit 10** for SVBFG's funding sources and those of the five largest US banks.) "Banks use a number of liabilities other than deposits to fund loans, including federal funds purchased, brokered deposits and borrowings from Federal Home Loan Banks," wrote the senior executive and the visiting scholar at the St. Louis Federal Reserve. "These all tend to be more expensive than deposits."[47]

Ultimately deposits were the lifeblood of a commercial bank and the primary source of funds for lending. Consequently, the more deposits the better. That said, excess deposits had always been a large concern at SVB.[48] "Every other bank is looking for deposits," said former SVB CEO Ken Wilcox in 2008. "They're dying to get their hands on deposits. We've got tons of them. Can you imagine what a problem that is? Because you've got to do something with them. You can't just sit on them."[49]

SVB leadership took the influx of deposits, and with loan growth slowing "amid exuberance in venture markets,"[50] it invested in long-term government bonds and mortgage-backed securities (MBSs), while largely classifying those investments using hold-to-maturity (HTM) accounting. (See **Exhibit 11**.) MBSs were debt obligations that represented claims to the cash flows (the principal and interest payments made by borrowers) from pools of mortgage loans purchased from banks, mortgage companies, and other originators.[51] With HTM accounting, SVB did not need to report on its balance sheet the current market value of its investments, which had dropped with the increase in interest rates. "Banks have to carry HTM instruments at amortized cost, or an adjusted version of the original price they paid," wrote a *Wall Street Journal* reporter. "Bonds the banks plan to sell need to be classified as available-for-sale securities and accounted for at fair market value. If banks sell any HTM securities, they must reclassify all of their HTM securities as available for sale and potentially take a big loss on the securities they didn't sell."[52] According to one scholar,

> The rationale for HTM accounting is that if a bank holds a security until its maturity, then the short-term dislocations between the book value and market value of the securities eventually wash out. The

[43] https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf.

[44] Brian Chappatta and Bloomberg, "How Silicon Valley Bank's Fate Was Sealed at the Moment of Its Triumph Just a Few Years Ago," *Fortune*, March 10, 2023, https://fortune.com/2023/03/10/how-silicon-valley-banks-fate-sealed-2021-venture-capital-bonds-interest-rates/ (accessed June 7, 2023).

[45] Generally, FDIC-insured banks paid a quarterly fee to the FDIC, which went to the agency's deposit insurance fund.

[46] https://www.npr.org/2023/03/15/1163269781/silicon-valley-bank-svb-collapse-history.

[47] Christopher J. Neely and Michelle Clark Neely, "Interest Rate Risk, Bank Runs and Silicon Valley Bank," St. Louis Federal Reserve, May 11, 2023, https://www.stlouisfed.org/publications/regional-economist/2023/may/interest-rate-risk-bank-runs (accessed June 30, 2023).

[48] https://www.vice.com/en/article/pkayyg/how-silicon-valley-bank-made-a-business-out-of-making-vcs-happy.

[49] https://www.vice.com/en/article/pkayyg/how-silicon-valley-bank-made-a-business-out-of-making-vcs-happy.

[50] Jenny Surane, "SVB's Giant Treasury Book, Tech Focus Made It Uniquely Shaky (1)," Bloomberg Law, March 10, 2023, https://news.bloomberglaw.com/mergers-and-acquisitions/svbs-giant-treasury-book-tech-focus-made-it-uniquely-shaky (accessed June 7, 2023).

[51] "Mortgage-Backed Securities and Collateralized Mortgage Obligations," US Securities & Exchange Commission, https://www.investor.gov/introduction-investing/investing-basics/glossary/mortgage-backed-securities-and-collateralized (accessed June 14, 2023).

[52] https://www.wsj.com/articles/banks-investors-revive-push-for-changes-to-securities-accounting-after-svb-collapse-99caa9ce.

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> critical condition for using HTM accounting is, therefore, that banks declare that they have both the intent and the ability to hold the securities until they mature such that the bank does not have to close the position at a significant accounting loss. Otherwise, banks must value their securities using available for sale (AFS) accounting, which forces them to mark the securities in their balance sheets using current market prices and to recognize unrealized losses on those securities in their statements of comprehensive income.[53]

The use of HTM accounting on the majority of its investments allowed SVB to exclude unrealized losses on those investments from equity write-downs, provided that SVB did not sell them.

The government bonds that SVB planned to hold to maturity were still useful collateral, even if they were declining in value. "Because it's high quality, we can borrow against it for liquidity purposes," said SVBFG CEO and president Greg Becker. "We have no intention of selling it as we can borrow against it."[54]

A Technology-Fueled Bank Run

According to one report, Moody's Investor Services (Moody's) had contacted SVB leaders in the days before the bank's collapse to inform them that Moody's was going to downgrade the bank's credit rating. Anonymous sources familiar with the situation said the downgrade was in response to the loss in value of the bonds, which SVB had largely bought before March 2022, due to higher interest rates. (See **Exhibit 12** for yields on 10-year Treasury bonds.) Yields on bonds and securitized investments issued after SVBFG had purchased them meant the value of its investments was less than what it had paid. *Financial Times* reporters wrote,

> Searching for yield in an era of ultra-low interest rates, [SVB] ramped up investment in a $120bn portfolio of highly rated government-backed securities, $91bn of these in fixed-rate mortgage bonds carrying an average interest rate of just 1.64 per cent. While slightly higher than the meagre returns it could earn from short-term government debt, the investments locked the cash away for more than a decade and exposed it to losses if interest rates rose quickly.[55]

Worried the downgrade could undermine the confidence of investors and clients in the bank's financial health, SVBFG leadership met with Goldman Sachs Group, Inc., (Goldman Sachs) for advice.[56] On March 8, 2023, SVBFG announced that it had commenced an underwritten public offering to raise $2.25 billion in common and preferred shares and that it had sold to Goldman Sachs $21 billion of AFS securities, mostly US Treasuries, with a yield of 1.79% and a duration of 3.6 years, at an after-tax loss of $1.8 billion.[57] "As an adviser

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[53] João Granja, "Bank Fragility and Reclassification of Securities into HTM" (working paper, Becker Friedman Institute for Economics, University of Chicago, April 25, 2023), https://bfi.uchicago.edu/working-paper/bank-fragility-and-reclassification-of-securities-into-htm/ (accessed June 28, 2023). "The statement of comprehensive income is a financial statement that summarizes both standard net income and other comprehensive income (OCI)…OCI consists of all unrealized gains and losses on assets that are not reflected in the income statement," CFI Team, "Statement of Comprehensive Income," Corporate Finance Institute (CFI), May 7, 2023, https://corporatefinanceinstitute.com/resources/accounting/statement-of-comprehensive-income/ (accessed July 24, 2023).

[54] Mark Vandevelde, Antoine Gara, Joshua Franklin, Colby Smith, and Tabby Kinder, "Silicon Valley Bank: The Multiple Warning Signs That Were Missed," *Financial Times*, April 25, 2023, (accessed via Factiva, June 30, 2023).

[55] Tabby Kinder, Antoine Gara, Joshua Franklin, and George Hammond, "Silicon Valley Bank: The Spectacular Unravelling of the Tech Industry's Banker," *Financial Times*, March 12, 2023, https://www.ft.com/content/b556badb-8e98-42fa-b88e-6e7e0ca758b8 (accessed June 28, 2023).

[56] "Silicon Valley Bank's Demise Began with a Downgrade Threat," Reuters, March 11, 2023, https://www.cnbc.com/2023/03/11/silicon-valley-banks-demise-began-with-downgrade-threat.html (accessed June 20, 2023).

[57] "SVB Financial Group Announces Proposed Offerings of Common Stock and Mandatory Convertible Preferred Stock," SVB Financial Group press release, March 8, 2023, https://ir.svb.com/news-and-research/news/news-details/2023/SVB-Financial-Group-Announces-Proposed-Offerings-of-Common-Stock-and-Mandatory-Convertible-Preferred-Stock/default.aspx (accessed June 16, 2023); "Strategic Actions/Q1'23 Mid-Quarter Update," SVB Financial Group, March 8, 2023, https://www.sec.gov/Archives/edgar/data/719739/000119312523064680/d430920dex992.htm (accessed July 7, 2023); Echo Wang, Niket Nishant, and Saeed Azhar, "SVB Says Goldman Sachs Was Buyer of Portfolio It Booked Losses on," Reuters, March 14, 2023, https://www.reuters.com/business/finance/goldman-sachs-bought-svbs-bond-portfolio-lender-says-2023-03-14/ (accessed July 18, 2023).

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to Silicon Valley Bank, Goldman Sachs…tried to pull off a last-minute capital raise to save the firm from collapse," *New York Times* reporters wrote. "The two-part plan of raising capital and buying debt couldn't save Silicon Valley Bank. Meanwhile, the compensation Goldman collected and how it managed the relationship with the bank could raise new questions."[58] By the end of the day, SVBFG's share price dropped 60%. Consequently, PE firm General Atlantic and other investors walked away from any equity investments.

"This was a classic asset-liability mismatch, triggered by higher rates, and compounded by leverage," said a senior executive at Fidelity Investments.[59] (See **Appendix** for more on maturity mismatch. **Exhibit 13** compares book value with fair-market value of SVBFG's balance sheet on December 31, 2022.)

The news spooked SVB's clients, leading to a bank run. On March 9, 2023, clients withdrew $42 billion, nearly 25% of the bank's total deposits, within 10 hours.[60] Historically, bank run withdrawals in one day were 10% or less of total deposits.[61] As such, the regulatory framework established by the Basel Committee on Banking Supervision only required banks to ensure they had protection against withdrawals of 10% or less in one day.[62] Peter Thiel's VC firm, Founders Fund, withdrew all its money, which some said sparked the run,[63] and advised its portfolio companies to make withdrawals.[64] CEO Becker pleaded with clients, saying, "If everybody is telling each other that SVB is in trouble, that will be a challenge…stay calm. That's my ask. We've been there for 40 years, supporting you, supporting the portfolio companies, supporting venture capitalists."[65]

The run was hastened by several factors—technology, SVB's concentration of clients in the VC orbit, and the high percentage of uninsured deposits. Some companies were concerned about making their payroll. At a later hearing, Becker told the US House Financial Services Subcommittee on Financial Institutions and Monetary Policy,

> As the bank run was ongoing, we were working to access additional liquidity when I was informed the morning of March 10 that the FDIC would be taking possession of SVB. That day, another roughly $100 billion in deposits were requested to be withdrawn, bringing the total actual and requested deposit outflow to roughly $142 billion, or about 80 percent of total deposits, over two days. To put the unprecedented velocity of this bank run in context, the previous largest bank run in U.S. history was $19 billion in deposits over the course of 16 days.[66]

The use of social media to get the word out on what was happening at SVB and the ability to withdraw deposits online (e.g., using a mobile phone) fueled the run. "While many internal factors made SVB susceptible to a bank run, both social media and digital technology accelerated the volume and speed of the deposit

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[58] Andrew Ross Sorkin, Ravi Mattu, Bernhard Warner, Sarah Kessler, Michael J. de la Merced, Lauren Hirsch, and Ephrat Livni, "Goldman Sachs Eyes a Big Payout from Silicon Valley Bank Deal," *New York Times*, March 15, 2023 (accessed via Factiva, July 18, 2023).

[59] "Global Banking System Turmoil," Fidelity Viewpoints, March 21, 2023, https://www.fidelity.com/learning-center/personal-finance/silicon-valley-bank-collapse (accessed June 30, 2023).

[60] *Joint Hearing Entitled: Continued Oversight over Regional Bank Failures, before the Subcommittees on Financial Institutions and Monetary Policy and Oversight and Investigations*, 118th Congress, May 17, 2023 (testimony by Greg Becker, former chief executive officer, Silicon Valley Bank) (accessed via Factiva, June 30, 2023).

[61] Gillian Tett, "Stress Testing Needs to Reverse to Move Forward," *Financial Times*, June 22, 2023, https://www.ft.com/content/6cd19c90-2c83-45cc-867a-d52fa19a6b92 (accessed June 28, 2023).

[62] https://www.ft.com/content/6cd19c90-2c83-45cc-867a-d52fa19a6b92.

[63] Harriet Alexander, "Peter Thiel Reveals He Had $50M in SVB When It Went Under—Despite Telling Investors to Pull Their Money Out of the Troubled Bank Hours before Its Collapse," *Daily Mail*, March 16, 2023 (accessed via Factiva, June 30, 2023).

[64] George Hammond, "Peter Thiel Had $50mn in Silicon Valley Bank When It Went Under," *Financial Times*, March 16, 2023 (accessed via Factiva, June 30, 2023).

[65] Alex Wilhelm and Natasha Mascarenhas, "Silicon Valley Bank's Shares Are Tanking as a Mess Unfolds," TechCrunch, March 9, 2023, https://techcrunch.com/2023/03/09/silicon-valley-banks-shares-are-tanking-as-a-mess-unfolds/ (accessed July 7, 2023).

[66] *Continued Oversight over Regional Bank Failures* (testimony by Greg Becker).

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outflows," concluded the California Department of Financial Protection and Innovation.[67] "This was the first Twitter fueled bank run," wrote the chair of the US House Financial Services Committee.[68] According to one study, tweets made by start-ups that referred to SVB went from a handful prior to the market opening on March 9 to well over 1,000 over the next 24 hours.[69] Scholars who examined what happened concluded,

> The failure of SVB was preceded by a large spike of public communication on Twitter by apparent depositors in SVB who used the forum to discuss the trouble the bank was facing, and more importantly, their intentions to withdraw their deposits from SVB. The openness and speed of this coordination around a bank run is unprecedented.[70]

SVB's concentration on VCs and the start-ups they funded also played a role in the bank run. Some VC firms instructed their portfolio companies to withdraw their money from SVB. "When VCs started to hear about potential issues, they weren't just worried about one account, startup, or fund, but everything they had. To the VCs, any risk to SVB is a risk to their entire portfolio," said one expert. "So the SVB deposit base was a lot less stable than it looked."[71]

The percentage of SVB's total deposits that were not insured by the FDIC—around 94%—was far greater than typically seen in the banking industry (see **Exhibit 14**). According to one estimate, the bank had around 37,000 clients, and the average deposit size was over $4 million.[72] Streaming platform Roku had nearly $500 million, or 26% of its cash and cash equivalents, at SVB.[73] The average deposit size in the United States was around $40,000.[74]

On March 10, 2023, the California Department of Financial Protection and Innovation (DFPI) announced that it had taken possession of SVB, "citing inadequate liquidity and insolvency."[75] The DFPI appointed the FDIC as receiver of SVB. Moody's chopped its credit ratings on SVB's long-term local currency bank deposit to Caa2 from A1 and dropped its issuer ratings from Baa1 to C.[76] S&P Global Ratings also slashed its issuer credit rating on SVB to a junk grade of D from an investment grade of BBB and lowered its rating on SVBFG to CC from BBB− (the lowest rating for an investment-grade bond).[77]

On March 26, the FDIC announced that First Citizens Bank & Trust (First Citizens), the 30th largest US bank, was acquiring parts of SVB, including about $72 billion (in loans) of SVB's approximate $167 billion of

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[67] "Review of DFPI's Oversight and Regulation of Silicon Valley Bank," Department of Financial Protection and Innovation (DFPI), May 8, 2023, https://dfpi.ca.gov/wp-content/uploads/sites/337/2023/05/Review-of-DFPIs-Oversight-and-Regulation-of-Silicon-Valley-Bank.pdf (accessed June 15, 2023).

[68] "McHenry Statement on Regulator Actions Regarding Silicon Valley Bank," US Financial Services Committee press release, March 12, 2023, https://financialservices.house.gov/news/documentsingle.aspx?DocumentID=408652 (accessed June 9, 2023).

[69] J. Anthony Cookson, Corbin Fox, Javier Gil-Bazo, Juan Felipe Imbet, and Christoph Schiller, "Social Media as a Bank Run Catalyst" (working paper no. 4422754, Université Paris-Dauphine, April 18, 2023), https://ssrn.com/abstract=4422754 or http://dx.doi.org/10.2139/ssrn.4422754 (accessed July 7, 2023).

[70] http://dx.doi.org/10.2139/ssrn.4422754.

[71] https://www.vice.com/en/article/pkayyg/how-silicon-valley-bank-made-a-business-out-of-making-vcs-happy.

[72] "The 5 Causes That Caused the Bankruptcy of Silicon Valley," 24 World Hours, March 12, 2023, https://24hoursworlds.com/market/383953/ (accessed Oct. 11, 2023).

[73] Megan Cerullo, "These Are Some of the Companies Impacted by the Silicon Valley Bank Collapse," CBS News, March 13, 2023, https://www.cbsnews.com/news/bank-collapse-silicon-valley-bank-customers-roblox/ (accessed July 9, 2023).

[74] John Cunnison, "Silicon Valley Bank and Its Demise: What You Should Know," Baker Boyer, March 13, 2023, https://www.bakerboyer.com/resources/articles/silicon-valley-bank-and-its-demise-what-you-should-know (accessed June 16, 2023).

[75] "California Financial Regulator Takes Possession of Silicon Valley Bank," DFPI press release, March 10, 2023, https://dfpi.ca.gov/2023/03/10/california-financial-regulator-takes-possession-of-silicon-valley-bank/ (accessed June 20, 2023).

[76] "Moody's Downgrades Credit Ratings on Silicon Valley Bank, SVB Financial," Reuters, March 10, 2023, https://money.usnews.com/investing/news/articles/2023-03-10/moodys-downgrades-credit-ratings-on-silicon-valley-bank-svb-financial (accessed June 20, 2023).

[77] "Research Update: Silicon Valley Bank Rating Lowered to 'D' And SVB Financial Group Rating Lowered to 'CC'; Ratings Subsequently Withdrawn," S&P Global, March 10, 2023, https://www.spglobal.com/ratings/en/research/articles/230310-research-update-silicon-valley-bank-rating-lowered-to-d-and-svb-financial-group-rating-lowered-to-cc-rat-12666691 (accessed June 20, 2023).

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total assets at a discount, paying $16.5 billion.[78] The remaining securities would stay in receivership. First Citizens would handle SVB's $56 billion deposits. (See **Exhibit 15** for timeline of key events leading up to First Citizens' acquisition.)

The Aftermath

Contagion

On March 12, 2023, two days after SVB's collapse, regulators took over Signature Bank of New York (SBNY). The FDIC reported that SBNY's failure was primarily because of "illiquidity by contagion."[79] SVB was not the only factor in SBNY's failure: just two days before SVB's collapse, Silvergate Bank, based in Southern California, had announced it was self-liquidating. Silvergate and SBNY both served cryptocurrency companies. Silvergate saw those companies pull out deposits after the fall of FTX, the cryptocurrency exchange.

SBNY's uninsured deposits ranged from 63% to 82% of total assets during the FDIC's period of review.[80] "SBNY's overreliance on this funding source was the primary driver of the bank's elevated liquidity risk profile," wrote the FDIC. "However, SBNY did not sufficiently establish policies and controls to address this key risk."[81] In addition, SBNY had expanded into the digital asset markets, which "exposed SBNY to greater susceptibility to liquidity, reputation, and regulatory risk due to the uncertainty and volatility of the digital asset space,"[82] and made it vulnerable "to bank runs and contagion, particularly in regards to crypto-related entities such as FTX, Alameda, and Silvergate…[and] increased the volatility and susceptibility of SBNY's more traditional depositor sources to event shocks and depositor runs."[83]

Federal regulators and regulations

On March 12, the US Treasury secretary, after consulting with US president Joe Biden, approved a systemic risk exception authorizing the FDIC to guarantee the uninsured deposits at SVB and SBNY. Both the FDIC and the Fed believed that not guaranteeing the uninsured deposits would likely lead to more bank runs, with which the secretary concurred. The Fed created the Bank Term Funding Program to provide eligible banks with additional liquidity by allowing the 12 Reserve Banks to provide loans of up to one year.[84]

In its review of the collapses at SVB and SBNY, the US General Accounting Office concluded that in the "5 years prior to 2023, regulators identified concerns with Silicon Valley Bank and Signature Bank, but both banks were slow to mitigate the problems the regulators identified, and regulators did not escalate supervisory actions in time to prevent the failures."[85]

In April 2023, the Fed released its own findings on SVB:

[78] Stan Cboe and Associated Press, "Silicon Valley Bank's Sale to First Citizens Has Boosted Trust in First Republic and Other Regional Banks—for Now," *Fortune*, March 28, 2023, https://fortune.com/2023/03/28/silicon-valley-bank-sale-first-citizens-boosted-trust-first-republic-pacwest-regional-banks-for-now/ (accessed July 9, 2023).

[79] "FDIC'S Supervision of Signature Bank," FDIC, April 28, 2023, https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf (accessed June 20, 2023).

[80] https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf.

[81] https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf.

[82] https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf.

[83] https://www.fdic.gov/news/press-releases/2023/pr23033a.pdf.

[84] "Preliminary Review of Agency Actions Related to March 2023 Bank Failures," US Government Accountability Office Report to the Committee on Financial Services, House of Representatives, April 2023, https://www.gao.gov/assets/gao-23-106736.pdf (accessed June 21, 2023).

[85] https://www.gao.gov/assets/gao-23-106736.pdf.

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> [SVB] failed because of a textbook case of mismanagement by the bank. Its senior leadership failed to manage basic interest rate and liquidity risk. Its board of directors failed to oversee senior leadership and hold them accountable. And Federal Reserve supervisors failed to take forceful enough action…Regulatory standards for SVB were too low, the supervision of SVB did not work with sufficient force and urgency, and contagion from the firm's failure posed systemic consequences not contemplated by the Federal Reserve's tailoring framework.[86]

In addition, from April 2022 to January 2023, SVBFG was without a chief risk officer, a position that reported directly to the board's risk committee. (See **Exhibit 16** for board members.) Ken Wilcox, former SVBFG CEO, said,

> There are multitudinous lines of defense in a well-run bank…A decision like the one that brought down the bank…had nothing to do with lending, nothing to do with the business model, it was all about what do we do with all of those excess deposits…The decision to invest a significant portion of those excess deposits in longer term treasuries was the fatal decision… [Normally] this decision would be made by the CFO and the CEO should be looking at that and saying to the CFO, "you know that's not a good idea, we shouldn't do that." But if that didn't happen, there was a group called ALCO, [the] asset liability committee—senior people in the bank. ALCO should have caught it and said, "that's not a good idea." And if ALCO missed it, there is always the finance committee of the board, and they should have caught it…the finance committee of the board when I was there had at least a couple of people that would have caught that.[87]

In the wake of the SVB and SBNY closures, 49 Democratic lawmakers and one Independent introduced a bill (the Secure Viable Banking Act, or SVB Act) calling for the repeal of provisions made in 2018 to the Dodd-Frank Wall Street Reform and Consumer Protection Act of 2010 (Dodd-Frank). Dodd-Frank, established to address issues that had surfaced during the 2008 financial crisis, originally required banks with assets of $50 billion or more to undergo risk-management oversight such as stress testing and higher liquidity requirements. The 2018 provisions had raised to $250 billion the minimum assets that caused a bank to be subject to certain rigorous risk-management oversight. In other words, banks considered "too big to fail," meaning those whose failure would create systemic risk to the stability of the US banking and financial systems, were those with assets of $250 billion or more. The SVB Act sought to restore the asset requirement levels to $50 billion.

Stress testing

The Fed conducted stress tests to ensure that large banks (those with $100 billion or more of assets) were sufficiently capitalized and able to lend to households and businesses even in a severe recession. Stress tests evaluated the financial resilience of banks by estimating losses, revenues, expenses, and resulting capital levels under a minimum of two Fed-developed hypothetical recession scenarios. The Fed conducted annual stress tests, using data provided by the banks, to test a bank's capital adequacy during times of stress, and it publicly disclosed bank-level results. Banks were also required to conduct and publicly disclose the results of their company-run stress tests based on their risk profiles.[88] The Fed's test scenarios, however, did not consider scenarios in which interest rates increased as quickly or as rapidly as they did.

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[86] https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf.

[87] "300. Leadership through Culture at SVB feat. Ken Wilcox," YouTube video, 1:12:00, posted by "unSILOed Podcast with Greg LaBlanc," July 3, 2023, https://www.youtube.com/watch?v=cbPVYExov8c (accessed July 8, 2023).

[88] "Stress Tests," Board of Governors of the Federal Reserve System, June 22, 2022, https://www.federalreserve.gov/supervisionreg/stress-tests-capital-planning.htm (accessed July 8, 2023).

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In June 2021, SVBFG breached $100 billion in average total consolidated assets, meaning that it would now be subject to Category IV, a higher level of capital, liquidity, and risk-management requirements. "However, at the time of its failure, an important subset of Category IV capital and liquidity requirements, including supervisory stress testing, the stress capital buffer, the liquidity coverage ratio (LCR), and the net stable funding ratio (NSFR),[89] were not yet applied to SVBFG because of applicable transition periods in the rules," concluded the Fed. The Fed also concluded that SVB repeatedly failed its own internal stress testing and that SVBFG had 31 open supervisory findings when it failed in March 2023, about triple the number at peer firms; and SVB's supervisory findings included core areas, such as governance and risk management, liquidity, interest-rate risk management, and technology. The Fed also blamed itself, writing,

> [Our] oversight of SVBFG proved inadequate for the well-documented and significant vulnerabilities and managerial weaknesses at SVBFG. The record shows that supervisors identified some of the material issues, but also underappreciated important ones, particularly during the period of SVBFG's rapid growth…SVB's foundational problems were widespread and well-known, yet core issues were not resolved, and stronger oversight was not put in place. After risks were identified, supervisors did not take sufficient steps to ensure that SVBFG fixed them in a timely fashion. This reflects a complex combination of many factors within the Federal Reserve, including delays in applying more stringent standards as SVBFG grew rapidly, the resources devoted to SVBFG supervision, an approach that emphasized consensus and the continued accumulation of evidence even as SVBFG deteriorated, and a shift in the stance of supervision policy that was amplified by the COVID-19 pandemic.[90]

In the wake of SVB's failure, the Fed's vice chair for supervision, Michael Barr, floated the idea of reverse stress testing: "Instead of thinking of a stressful scenario and then seeing how it would play out through, say, the balance sheet of a firm, you look at a bank and you say, well, what would it take to really break this institution?" In other words, Barr explained, "what are the different ways this institution could die?"[91] A *Financial Times* senior editor wrote,

> How the Fed—or the banks—might actually adopt this innovation in the future is unclear…That is partly because reverse stress tests would force financiers to become more imaginative about the future. This matters deeply right now given that markets are being hit by shocks that have not been seen for many decades (if ever), be that high inflation, protectionism, war or rapid digitization. The other big merit of reverse stress testing is that this process could also force financiers to think more about qualitative issues, alongside quantitative ones, when it comes to judging risk.[92]

Another bank failure

On May 1, 2023, regulators took over another bank—California-based First Republic—despite an infusion of $30 billion into the bank from 11 US banks in March. Those banks had effectively opened bank accounts at First Republic and made deposits in hopes of boosting confidence in the bank and the US banking industry more generally.[93] It was the second-largest bank failure in US history. First Republic catered to wealthy clients, and nearly 70% of its deposits were uninsured (refer to **Exhibit 14**). Clients, both high-net-worth individuals

[89] The NSFR was a ratio that had to equal or exceed 100%. It was calculated by dividing a bank's total available stable funding (ASF) by its total required stable funding (RSF). ASF was the portion of a bank's capital and liabilities that would remain with the bank for more than one year. RSF was the amount of stable funding that a bank was required to hold given the liquidity characteristics and residual maturities of its assets and the contingent liquidity risk arising from its off-balance sheet exposures. "Net Stable Funding Ratio (NSFR) – Executive Summary," Bank for International Settlements (BIS), June 28, 2018, https://www.bis.org/fsi/fsisummaries/nsfr.htm (accessed July 10, 2023).

[90] https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf.

[91] https://www.ft.com/content/6cd19c90-2c83-45cc-867a-d52fa19a6b92.

[92] https://www.ft.com/content/6cd19c90-2c83-45cc-867a-d52fa19a6b92.

[93] Steve Inskeep and David Gura, "The Biggest Banks in the U.S. Are Stepping in to Save First Republic Bank," NPR, March 17, 2023, https://www.npr.org/2023/03/17/1164146742/the-biggest-banks-in-the-u-s-are-stepping-in-to-save-first-republic-bank (accessed June 21, 2023).

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and companies, had withdrawn around $100 billion of their deposits after SVB's collapse and moved them to larger banks they felt were safer.[94] One bank analyst said First Republic was not known to have any big underlying problems but was "caught up in the selloff."[95] "First Republic saw a mass exodus of depositors to larger institutions, who feared that the bank would not have the capital to cover huge unrealized losses on its books due to rising interest rates," explained one reporter.[96] For $10.6 billion, JPMorgan Chase acquired $173 billion of First Republic's loans, $30 billion of its securities, and $92 billion of its deposits, and it agreed to repay the $25 billion of deposits large US banks had made in First Republic in March.[97] JPMorgan Chase did not assume First Republic's corporate debt or preferred stock. The FDIC agreed to backstop loan losses. All First Republic depositors became depositors of JPMorgan Chase Bank and had full access to all their deposits.

[94] Annie Gaus, Kevin Truong, and Matt Smith, "First Republic Bank Will Be Seized by Regulators, Reports Say," *San Francisco Standard*, April 28, 2023, https://sfstandard.com/business/first-republic-bank-will-be-seized-by-regulators-report-says/ (accessed June 21, 2023).

[95] https://www.npr.org/2023/03/17/1164146742/the-biggest-banks-in-the-u-s-are-stepping-in-to-save-first-republic-bank.

[96] Nicole Narea, "Is First Republic Bank's Failure Sign of a Slow-Motion Banking Crisis?" Vox, May 1, 2023, https://www.vox.com/business-and-finance/2023/5/1/23706910/first-republic-bank-failure-crisis-fdic (accessed June 21, 2023).

[97] "JPMorgan's Deal for Collapsed First Republic Explained," Reuters, May 1, 2023, https://www.reuters.com/business/finance/jpmorgans-deal-collapsed-first-republic-explained-2023-05-01/ (accessed July 24, 2023).

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Exhibit 1

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group Operating Segment Results, 2018 to 2022
(in millions of US dollars)

	2022	2021	2020	2019	2018
Revenues *(= net interest income − provision for credit losses + noninterest income)*					
Silicon Valley Bank	$4,948	$3,565	$2,428	$2,396	$2,063
SVB Private	493	270	96	53	64
SVB Securities	508	609	498	265	-
SVB Capital	(110)	487	226	122	101
Other items	(46)	863	529	375	323
Total revenues	$5,793	$5,794	$3,777	$3,211	$2,551
Net profit before tax					
Silicon Valley Bank	$3,391	$2,299	$1,417	$1,521	$1,270
SVB Private	132	47	41	13	39
SVB Securities	(95)	48	119	12	-
SVB Capital	(180)	416	175	91	78
Other items	(1,076)	(86)	(10)	(27)	(24)
Total net profit before tax	$2,172	$2,724	$1,742	$1,610	$1,363
Assets					
Silicon Valley Bank	$175,221	$140,362	$73,929	$56,043	$48,854
SVB Private	16,637	11,171	5,335	3,371	2,872
SVB Securities	936	830	557	398	-
SVB Capital	942	700	437	405	381
Other items	22,367	12,948	5,534	2,995	3,122
Total assets	$216,103	$166,011	$85,792	$63,212	$55,229
Deposits					
Silicon Valley Bank	NA	NA	NA	NA	NA
SVB Private	NA	NA	NA	NA	NA
SVB Securities	NA	NA	NA	NA	NA
SVB Capital	NA	NA	NA	NA	NA
Other items	NA	NA	NA	NA	NA
Total deposits	$175,378	$191,431	$103,194	$62,944	$49,992

Note: These figures may differ slightly from those reported on SVB Financial Group's SEC Form 10-K.

Data source: Capital IQ (accessed June 14, 2023).

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Exhibit 2

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group Consolidated Income Statement, 2020 to 2022
(in millions of US dollars, except per-share data)

	2022	2021	2020
Interest income:			
Loans	$3,208	$1,966	$1,520
Investment securities:			
Taxable	2,113	1,199	635
Nontaxable	140	106	61
Federal funds sold, securities purchased under agreements to resell, and other short-term investment securities	212	18	26
Total interest income	5,673	3,289	2,242
Interest expense:			
Deposits	862	62	60
Borrowings	326	48	25
Total interest expense	1,188	110	85
Net interest income	4,485	3,179	2,157
Provision for credit losses	420	123	220
Net interest income after provision for credit losses	4,065	3,056	1,937
Noninterest income:			
Gains (losses) on investment securities, net	(285)	761	421
Gains on equity warrant assets, net	148	560	237
Client investment fees	386	75	132
Wealth-management and trust fees	83	44	-
Foreign exchange fees	285	262	179
Credit card fees	150	131	98
Deposit service charges	126	112	90
Lending related fees	94	76	57
Letters of credit and standby letters of credit fees	57	51	47
Investment banking revenue	420	459	414
Commissions	98	79	67
Other	166	128	98
Total noninterest income	1,728	2,738	1,840
Total noninterest expense	3,621	3,070	2,035
Income before income tax expense	2,172	2,724	1,742
Income tax expense	563	651	448
Net income before noncontrolling interests and dividends	1,609	2,073	1,294
Net loss (income) attributable to noncontrolling interests and dividends	63	(240)	(86)
Preferred stock dividends	(163)	(63)	(17)
Net income available to common stockholders	$1,509	$1,770	$1,191
Earnings per common share (basic)	$25.58	$31.74	$23.05

Notes: The fiscal year ends December 31. Includes operating segments: SVB, SVB Private, SVB Capital, and SVB Securities, Inc.

Data source: SVB Financial Holdings SEC Form 10-K, 2022.

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Exhibit 3

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group Balance Sheet, 2020 to 2022
(in millions of US dollars, except par value and share data)

	Dec-31-2022	Dec-31-2021	Dec-31-2020
Assets:			
Cash and cash equivalents	$13,803	$14,586	$17,674
AFS securities at fair value	26,069	27,221	30,913
HTM securities at amortized cost net of credit losses	91,321	98,195	16,592
Nonmarketable and other equity securities	2,664	2,543	1,802
Total investment securities	120,054	127,959	49,307
Loans, amortized cost	74,250	66,276	45,181
Allowance for credit losses: loans	(636)	(422)	(448)
Net loans	73,614	65,854	44,733
Premises and equipment, net of depreciation and amortization	394	270	176
Goodwill	375	375	143
Other intangible assets, net	136	160	61
Lease right-of-use assets	335	313	210
Accrued interest receivable and other assets	3,082	1,791	3,206
Total assets	$211,793	$211,308	$115,511
Liabilities:			
Noninterest-bearing demand deposits	80,753	125,851	66,519
Interest-bearing deposits	92,356	63,352	35,463
Total deposits	173,109	189,203	101,982
Short-term borrowings	13,565	71	21
Lease liabilities	413	388	259
Other liabilities	3,041	2,467	3,972
Long-term debt	5,370	2,570	844
Total liabilities	195,498	194,699	107,078
SVB Financial Group stockholders' equity:			
Preferred stock	3,646	3,646	340
Common stock	-	-	-
Additional paid-in capital	5,318	5,157	1,585
Retained earnings	8,951	7,442	5,672
Accumulated other comprehensive income (loss)	(1,911)	(9)	623
Total stockholders' equity	16,004	16,236	8,220
Noncontrolling interests	291	373	213
Total equity	16,295	16,609	8,433
Total liabilities and equity	$211,793	$211,308	$115,511

Notes: AFS = available for sale; HTM = hold to maturity.

Data source: SVB Financial Holdings SEC Form 10-K, 2021 and 2022.

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Exhibit 4

The Sudden Implosion of Silicon Valley Bank

SVB Loan Portfolio Mix as of December 31, 2022

Credit Type	Percentage of Total Loans
Primarily private equity (PE) and venture capital (VC) capital call lines of credit secured by LP capital commitments.[a]	56%
Primarily low loan-to-value (LTV) mortgages to innovation economy influencers and legacy Boston Private[b] high-net-worth clients.	14%
Cash flow or balance sheet dependent loans to later- and corporate-stage innovation companies.	12%
Loans to mid- and later-stage innovation companies with revenues greater than $5 million.	6%
Loans to early-stage innovation companies with $0 to $5 million in revenues.	3%
Acquisition financing for commercial real estate (CRE) properties.	3%
Loans to facilitate PE sponsors' acquisition of businesses.	3%
Loans to wine producers and vineyard typically secured by high-quality real estate with low LTVs.	2%
Working capital revolving line of credit and term loans to non-innovation companies and nonprofits.	1%

[a] A capital call line of credit was a facility provided by SVB in exchange for interest. General partners (GPs) drew funds from those facilities often quarterly instead of calling capital directly from limited partners (LPs). A capital call (also known as a drawdown) was when a PE or VC firm requested a portion of the capital that its LPs had committed to invest, and used that "called" capital to make an investment or pay fees and expenses. "Cash Flow Management Component 3 of 4: Capital Call Lines of Credit," SVB, https://www.svb.com/emerging-manager-insights/starting-a-fund/cash-flow-management-lines-of-credit (accessed Oct. 30, 2023).
[b] SVB Financial Group acquired the private bank and wealth-management firm in 2021.

Source: Adapted from "SVB Q4 2022 Financial Highlights" January 19, 2023, https://s201.q4cdn.com/589201576/files/doc_financials/2022/q4/Q4_2022_IR_Presentation_vFINAL.pdf (accessed June 12, 2023).

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Exhibit 5

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group Metrics versus Select Banks of Similar Size and Silvergate Bank, 2018 to 2022

		2018	2019	2020	2021	2022
SVBFG	**Total assets*** (in billions)	$56.9	$71.0	$115.5	$211.3	$211.8
#16	**Net loans/total assets**	49.4%	46.3%	38.7%	31.2%	34.7%
	Profitability					
	Return on assets	1.9%	1.9%	1.4%	1.3%	0.8%
	Return on equity	21.1%	19.9%	17.2%	16.6%	9.8%
	Liquidity					
	Cash/assets	6.1%	8.8%	14.9%	6.3%	6.0%
	Net loans/total deposits	56.9%	53.2%	43.9%	34.8%	42.5%
	Asset quality					
	Nonperforming loans (NPL)/total assets	0.17%	0.15%	0.09%	0.04%	0.06%
	Price to tangible book value (TBV)*	2.05×	2.27×	2.76×	3.39×	1.20×
	Tier 1 capital ratio	13.6%	13.4%	11.9%	16.1%	15.4%
First Republic	**Total assets*** (in billions)	$99.2	$116.3	$142.5	$181.1	$212.6
#14	**Net loans/total assets**	76.0%	77.6%	78.5%	74.2%	78.1%
	Profitability					
	Return on assets	0.9%	0.9%	0.9%	0.9%	0.9%
	Return on equity	10.4%	10.0%	9.9%	10.7%	10.0%
	Liquidity					
	Cash/assets	2.8%	1.5%	3.6%	7.1%	2.0%
	Net loans/total deposits	95.4%	100.2%	97.4%	85.9%	94.1%
	Asset quality					
	NPL/total assets	0.05%	0.12%	0.13%	0.08%	0.05%
	Price to TBV*	1.98×	2.40×	2.67×	3.17×	1.68×
	Tier 1 capital ratio	11.7%	11.2%	11.2%	12.6%	11.6%
Citizens Bank	**Total assets*** (in billions)	$160.5	$165.7	$183.4	$188.4	$226.7
Financial Group	**Net loans/assets**	71.9%	71.1%	65.8%	67.1%	68.2%
#13	**Profitability**					
	Return on assets	1.1%	1.1%	0.6%	1.2%	1.0%
	Return on equity	8.4%	8.3%	4.7%	10.1%	8.8%
	Liquidity					
	Cash/assets	0.7%	0.8%	6.9%	4.9%	4.6%
	Net loans/total deposits	96.5%	94.0%	82.0%	81.9%	85.6%
	Price to TBV	1.11×	1.32×	1.14×	1.40×	1.50×
	Asset quality					
	NPL/total assets	0.48%	0.42%	0.56%	0.37%	0.42%
	Tier 1 capital ratio	11.3%	11.1%	11.3%	11.1%	11.1%
Signature Bank	**Total assets*** (in billions)	$47.4	$50.6	$73.9	$118.4	$110.4
of New York	**Net loans/total assets**	76.4%	76.9%	65.4%	54.4%	70.9%
#29	**Profitability**					
	Return on assets	1.1%	1.2%	0.8%	1.0%	1.2%
	Return on equity	12.0%	12.8%	10.0%	13.4%	16.9%
	Liquidity					
	Cash/assets	0.6%	0.6%	16.8%	25.0%	5.4%
	Net loans/total deposits	99.5%	96.2%	76.3%	60.7%	83.3%
	Price to TBV	1.32×	1.54×	1.43×	2.52×	0.93×
	Asset quality					
	NPL/Total Assets	0.23%	0.11%	0.16%	0.18%	0.17%
	Tier 1 capital ratio	12.1%	11.6%	11.2%	10.5%	11.2%
Silvergate Bank /	**Total assets*** (in billions)	$2.0	$2.1	$5.6	$16.0	$11.4
Silvergate Capital	**Net loans/total assets**	30.0%	28.6%	12.5%	5.6%	3.5%
Corp.	**Profitability**					
#506	Return on assets	1.1%	1.2%	0.7%	0.7%	(6.9%)
	Return on equity	16.9%	11.8%	9.9%	8.3%	(84.8%)
	Liquidity					
	Cash/assets	35.0%	4.8%	53.6%	33.8%	40.4%
	Net loans/total deposits	33.2%	36.6%	14.2%	6.2%	6.5%
	Asset quality					
	NPL/total assets	0.41%	0.28%	0.09%	0.03%	0.03%
	Price to TBV	NA	1.23x	4.89x	3.67x	0.48x
	Tier 1 capital ratio	25.0%	26.2%	22.9%	56.8%	57.1%

* As of December 31. Note: National rank based on total assets.

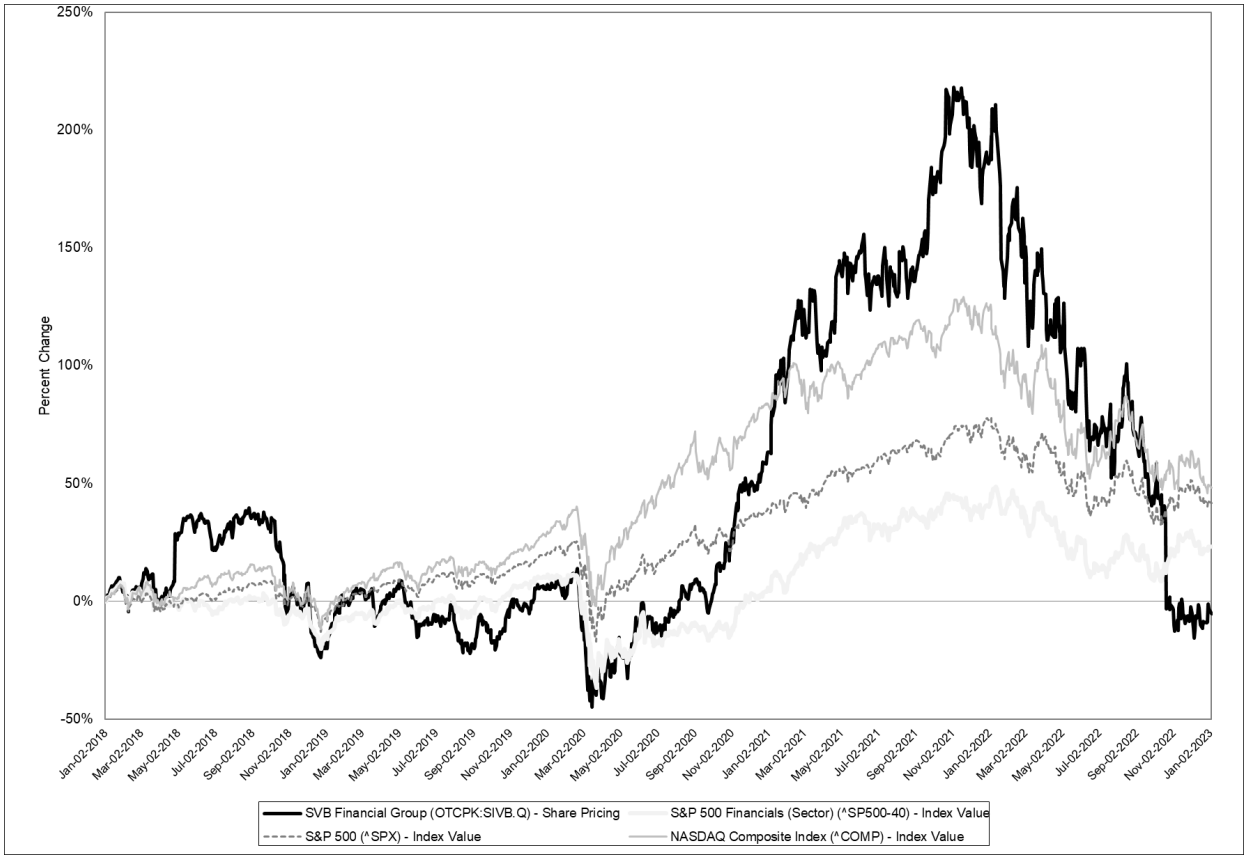
Sources: Capital IQ (accessed June 14 and 15, 2023) and author analysis.

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Exhibit 6

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group Stock Price versus Benchmarks, January 2, 2021, to January 3, 2023



Source: Created by authors using Capital IQ data (accessed June 28, 2023).

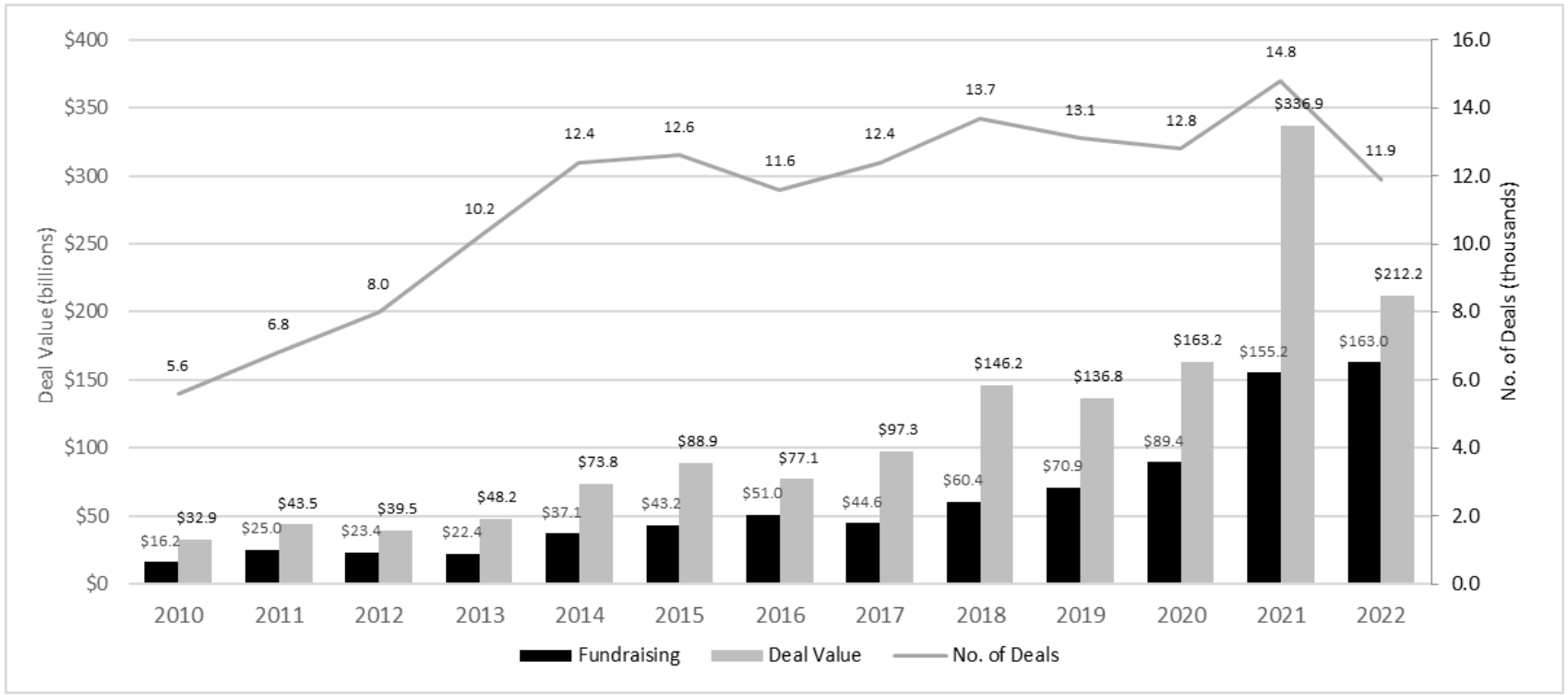
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Exhibit 7

The Sudden Implosion of Silicon Valley Bank

US Venture Capital Funds Raised and Investments in Venture Capital–Backed US Companies, 2010 to 2022



Note: Above shows equity financing into VC-backed US companies. Sources of cash investments included, but were not limited to, VC firms, corporate investors, other PE firms, and individuals. VC = venture capital; PE = private equity.

Data sources: "Deal Value ($b) by Year and Sector Group," Ernst & Young LLP, https://app.powerbi.com/view?r=eyJrIjoiNjk1Mzc4YzgtODMxOS00ODMxLWE1MTMtMjBiOGI5NmNhNTUzIiwidCI6IjViOTczZjk5LTc3ZGYtNGJlYi1iMjdkLWFhMGM3MGI4NDgyYyIsImMiOjh9 (accessed June 12, 2023), and "2023 Yearbook," National Venture Capital Association, https://nvca.org/wp-content/uploads/2023/03/NVCA-2023-Yearbook_FINALFINAL.pdf cited in "Venture Capital Fundraising in the United States from 2007 to 2022," Statista, https://www-statista-com.proxy1.library.virginia.edu/statistics/280260/venture-capital-fund-raising-by-holding-companies-in-the-usa/ (accessed June 16, 2023).

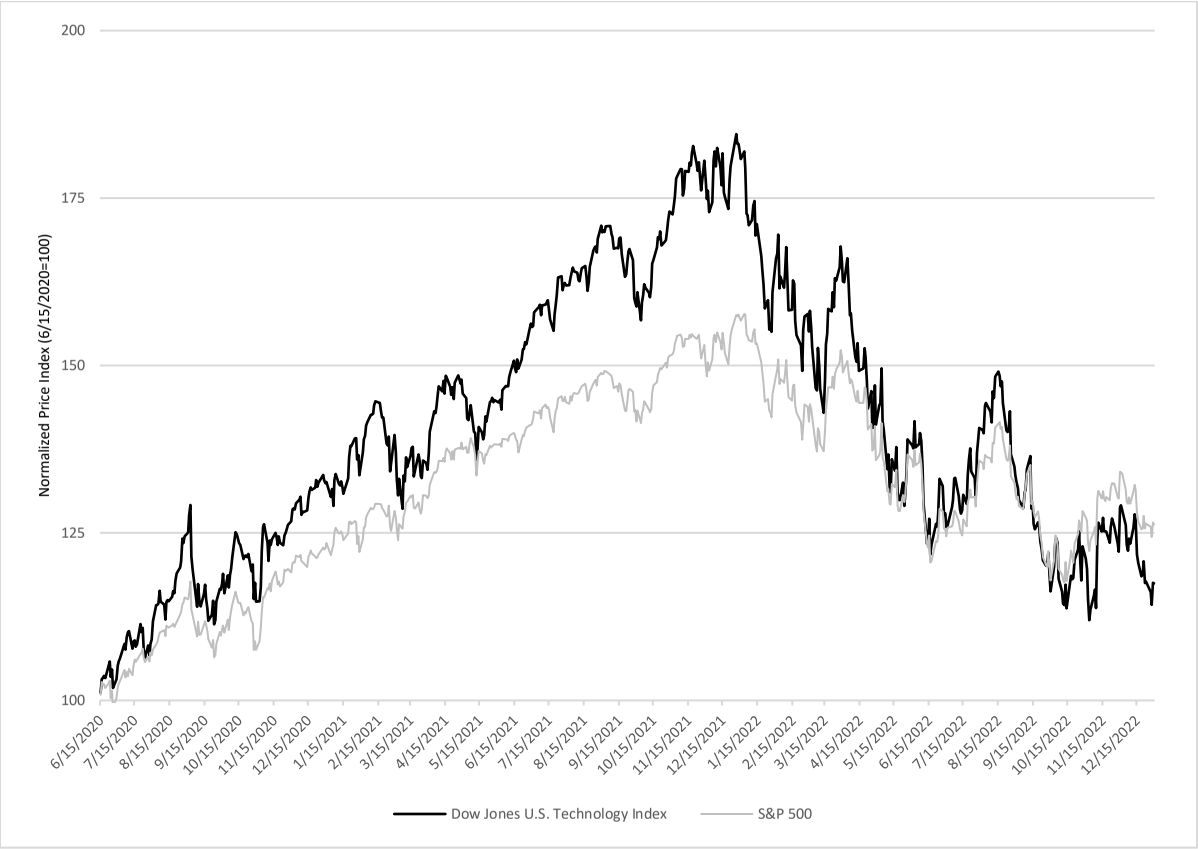
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Exhibit 8

The Sudden Implosion of Silicon Valley Bank

Dow Jones US Tech Sector Index versus S&P 500, June 15, 2020, to December 31, 2022



Source: "Dow Jones U.S. Technology Index," https://www.spglobal.com/spdji/en/indices/equity/dow-jones-us-technology-index/#overview (accessed June 16, 2023).

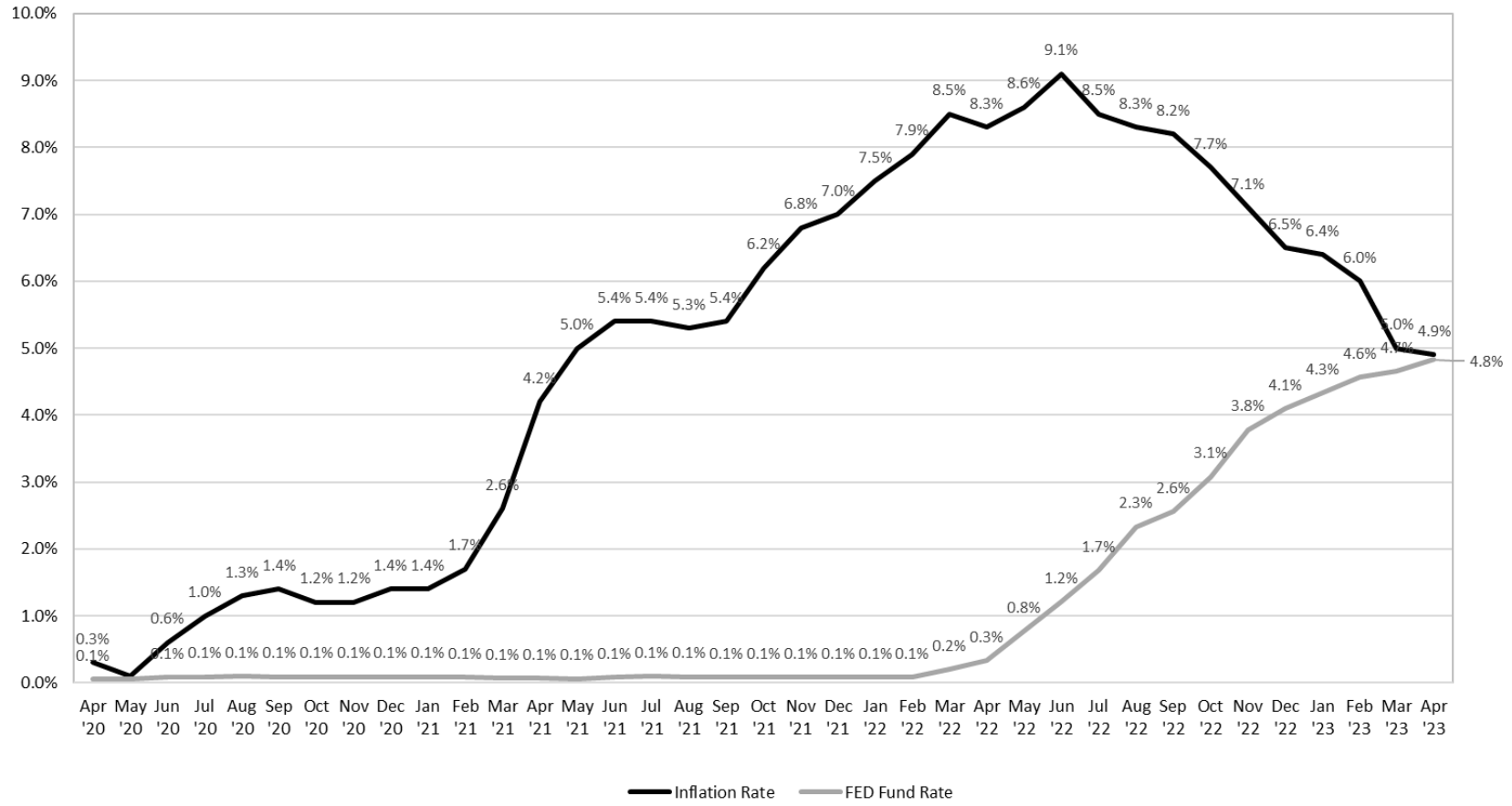
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Exhibit 9

The Sudden Implosion of Silicon Valley Bank

US Rate of Inflation and Federal Reserves' Funds Rate, Monthly, April 2020 to April 2023



Sources: Created by authors; data from "Monthly 12-Month Inflation Rate in the United States from April 2020 to April 2023," (graph; data from Bureau of Labor Statistics, May 10, 2023), Statista, https://www-statista-com.proxy1.library.virginia.edu/statistics/273418/unadjusted-monthly-inflation-rate-in-the-us/; and "Federal Funds Effective Rate," FRED Graph Observations, Federal Reserve Economic Data, https://fred.stlouisfed.org/series/FEDFUNDS (both accessed June 15, 2023).

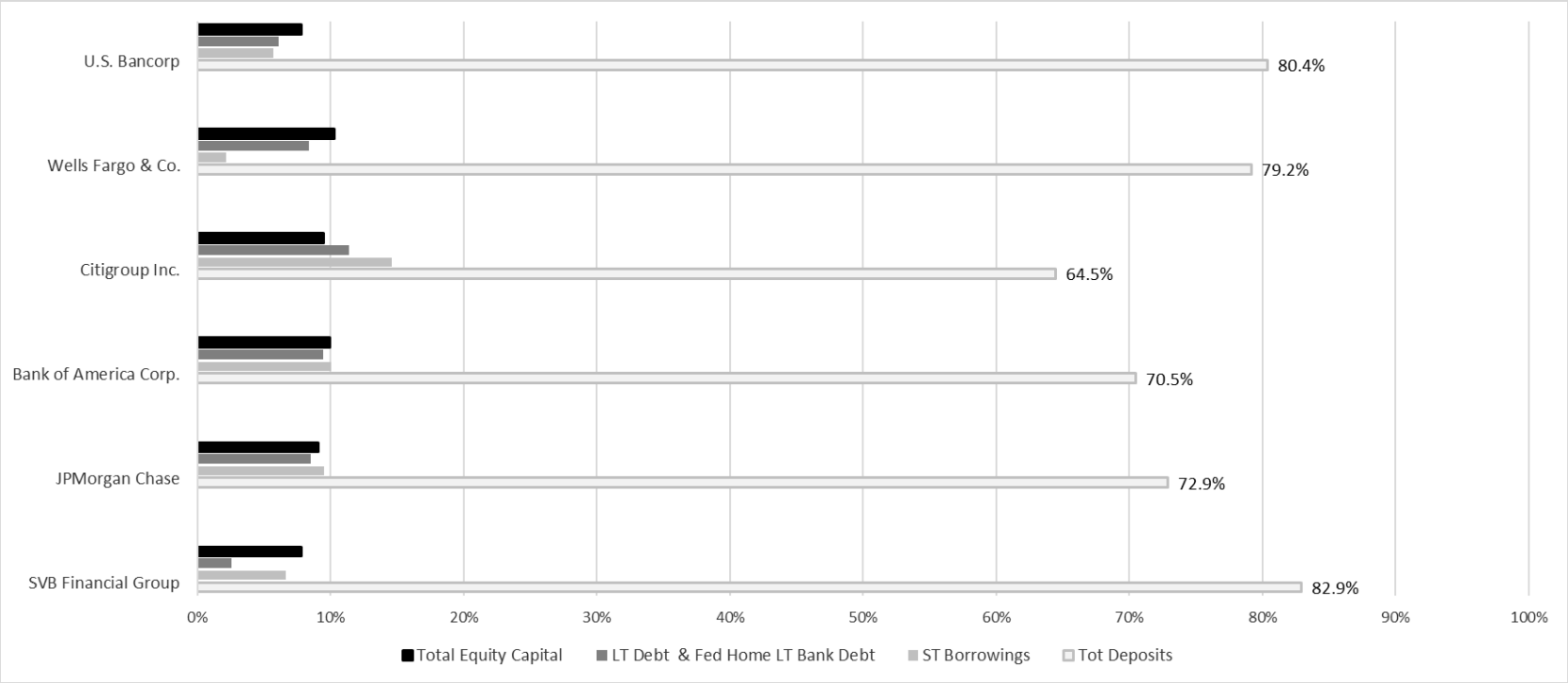
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Exhibit 10

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group and Five Largest US Banks—Funding Sources as of December 31, 2022



Note: Fed Home Long-Term (LT) Bank Debt as a percentage of total funding sources was 0% to 1% for the above banks. This represents secured loans provided to banks from a Federal Home Loan Bank, of which there were 11.

Source: Author analysis; data from Capital IQ (accessed June 29, 2023).

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Exhibit 11

The Sudden Implosion of Silicon Valley Bank

SVB Financial Group's Investment Securities, 2020 to 2022
(in millions of US dollars)

	As of December 31,		
	2022	**2021**	**2020**
AFS securities, at fair value:			
US Treasury securities	$16,135	$15,850	$4,470
US agency debentures	101	196	237
Foreign government debt securities	1,088	61	24
Residential MBS:			
Agency-issued MBS[a]	6,603	8,589	13,503
Agency-issued CMO-fixed rate	678	982	8,107
Agency-issued CMBS	1,464	1,543	4,572
Total AFS securities	26,069	27,221	30,913
Percentage of total investment securities	*21.7%*	*21.3%*	*62.7%*
HTM securities, at net carry value:			
US agency debentures	486	609	402
Residential MBS:			
Agency-issued MBS[a]	57,705	64,439	7,740
Agency-issued CMO-fixed rate	10,461	10,226	1,735
Agency-issued CMO-variable rate	79	100	137
Agency-issued CMBS	14,471	14,959	2,943
Municipal bonds and notes	7,416	7,156	3,635
Corporate bonds	703	706	-
Total HTM securities	91,321	98,195	16,592
Percentage of total investment securities	*76.1%*	*76.7%*	*33.7%*
Nonmarketable and other equity securities:			
Nonmarketable securities (fair value accounting)			
Consolidated VC and PE fund investments	147	130	89
Unconsolidated VC and PE fund investments	110	208	185
Other investments	183	164	61
Other equity securities in public companies (fair value accounting)	32	117	281
Nonmarketable securities (equity method accounting)			
VC and PE fund investments	605	671	362
Debt funds	5	5	5
Other investments	276	294	203
Investments in qualified affordable housing projects, net	1,306	954	616
Total nonmarketable and other equity securities	2,664	2,543	1,802
Percentage of total investment securities	*2.2%*	*2.0%*	*3.7%*
Total investment securities	$120,054	$127,959	$49,307

[a] Most MBS were issued by the Government National Mortgage Association (Ginnie Mae), a US government agency, or the Federal National Mortgage Association (Fannie Mae), and the Federal Home Loan Mortgage Corporation (Freddie Mac), US government-sponsored enterprises.

Notes: AFS = available for sale; HTM = hold to maturity; MBS = mortgage-backed security; CMBS = commercial mortgage-backed security; CMO = collateralized mortgage obligation; VC = venture capital; PE = private equity.

Source: SVB Financial Holdings SEC Form 10-K, 2022.

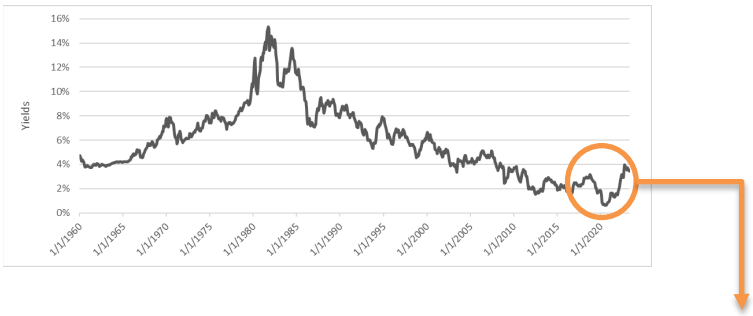
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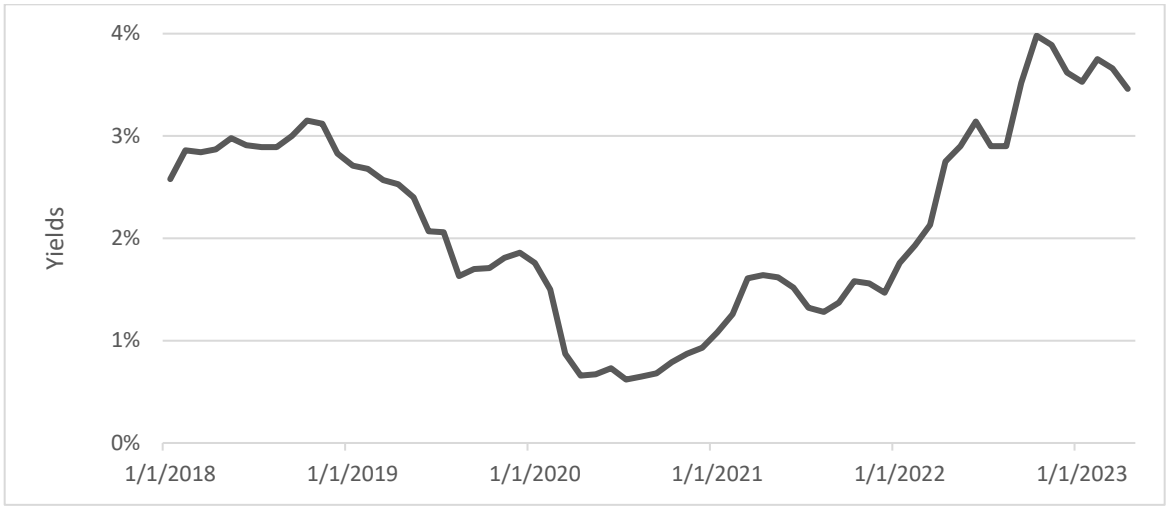
Exhibit 12

The Sudden Implosion of Silicon Valley Bank

Long-Term US Government 10-year Bond Yields, Monthly, January 1, 1960, to April 1, 2023



Long-Term US Government 10-year Bond Yields, Monthly, January 1, 2018, to April 1, 2023



Note: "SVB purchased most of its fixed-rate and longer-duration securities prior to the interest rate hikes the Fed started in March 2022. This is reflected in the yields of SVB's assets, which are lower than the current market rates. While the target federal funds rate was 4.25% to 4.50% as of 31 December 2022, the weighted average yield on the AFS securities was 1.56% and 1.66% for the HTM securities. In addition, about 40% of the AFS securities had maturities of more than five years and almost all the HTM securities had maturities of more than five years." Fatan Sabry, William Hrycay, and Sungi Lee, "What Happened to SVB? Not a 2008 Repeat Story," NERA Economic Consulting, March 24, 2023, https://www.nera.com/content/dam/nera/publications/2023/PUB_What_Happened_to_SVB_0323.pdf (accessed July 7, 2023).

Source: Adapted from Organisation for Economic Co-operation and Development, "Long-Term Government Bond Yields: 10-year: Main (Including Benchmark) for the United States [IRLTLT01USM156N]," FRED, Federal Reserve Bank of St. Louis, May 14, 2023, https://fred.stlouisfed.org/series/IRLTLT01USM156N (accessed May 15, 2023).

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Exhibit 13

The Sudden Implosion of Silicon Valley Bank

SVB Balance Sheet as of December 31, 2022: Book Value versus Fair Value

Book Value



Fair Value



Source: Created by authors using data in Faten Sabry, William Hrycay, and Sungi Lee, "What Happened to SVB? Not a 2008 Repeat Story," NERA Economic Consulting, March 24, 2023, https://www.nera.com/publications/archive/2023/what-happened-to-svb--not-a-2008-repeat-story.html#:~:text=The%20collapse%20of%20SVB% (accessed July 8, 2023).

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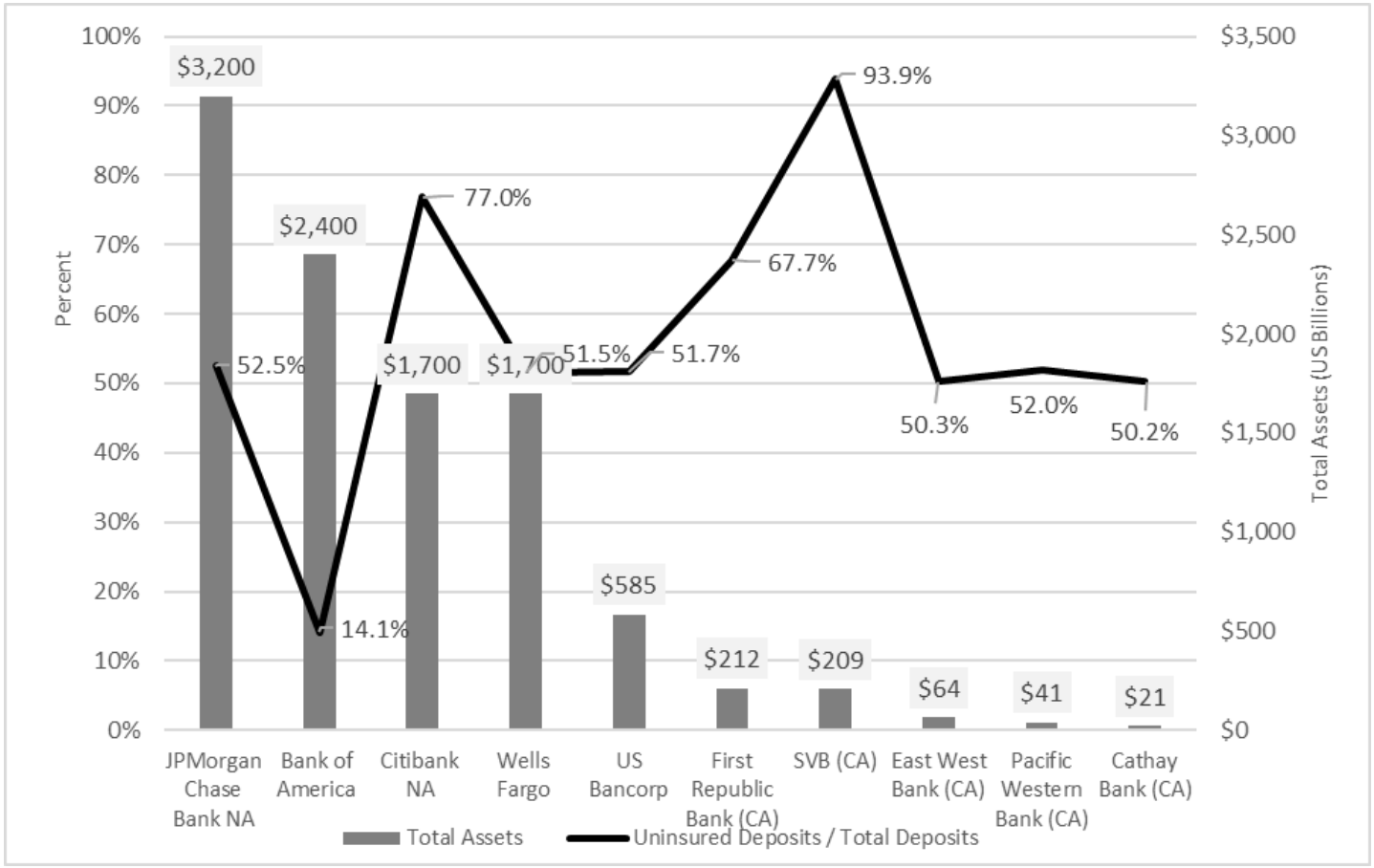
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Exhibit 14

The Sudden Implosion of Silicon Valley Bank

Uninsured Deposits as a Percentage of Total Deposits at Select US Banks (Largest US and Largest in California) as of December 31, 2022



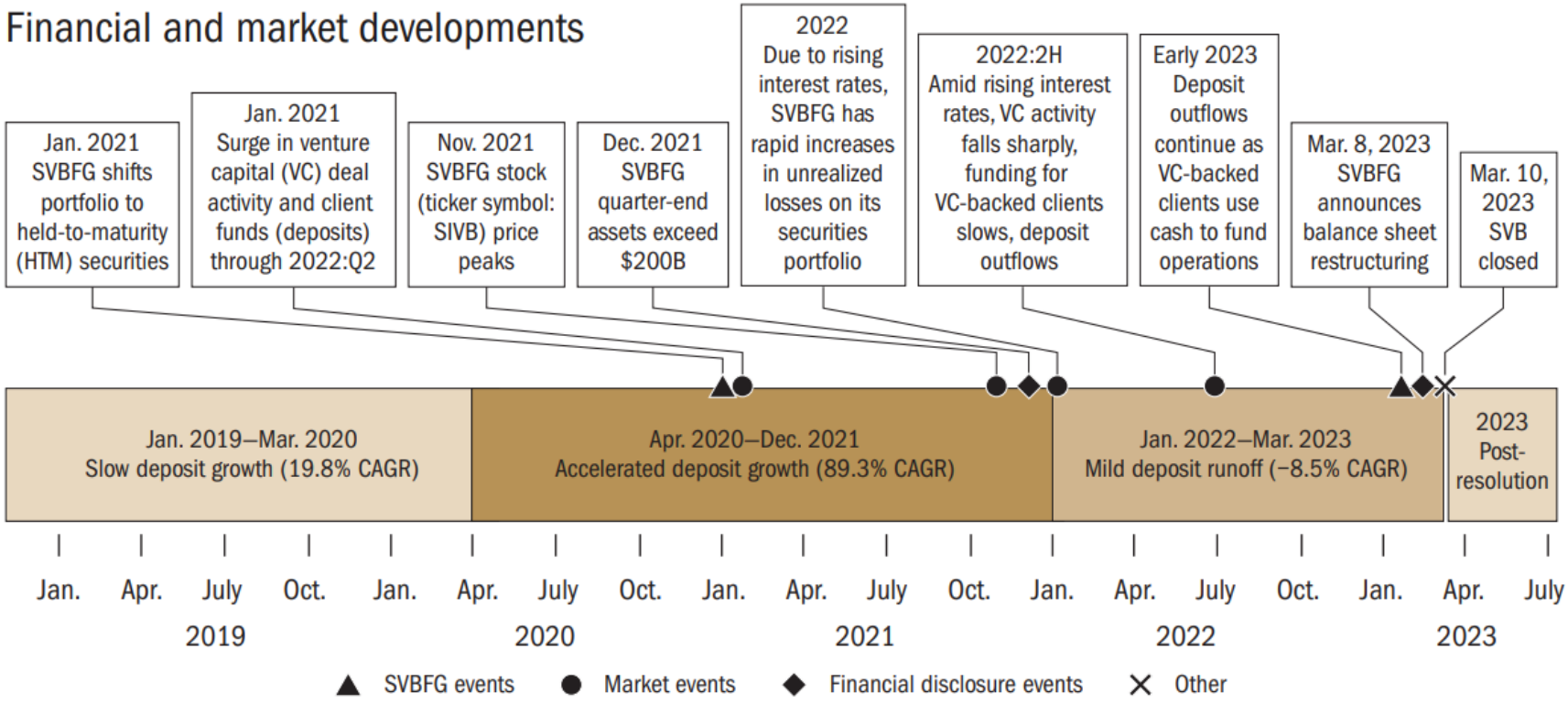
Source: Created by authors; data from "Review of DFPI's Oversight and Regulation of Silicon Valley Bank," Department of Financial Protection and Innovation (DFPI), May 8, 2023, https://dfpi.ca.gov/wp-content/uploads/sites/337/2023/05/Review-of-DFPIs-Oversight-and-Regulation-of-Silicon-Valley-Bank.pdf (accessed June 15, 2023).

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Exhibit 15

The Sudden Implosion of Silicon Valley Bank

Timeline: Key Developments, January 2019 to March 2023



Note: SVBFG = Silicon Valley Bank Financial Group.

Source: Michael S. Barr, "Re: Review of the Federal Reserve's Supervision and Regulation of Silicon Valley Bank," Board of Governors of the Federal Reserve System, April 28, 2023, https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf (accessed June 15, 2023).

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Exhibit 16

The Sudden Implosion of Silicon Valley Bank

SVB Financial Holdings Board of Directors, 2022

				Committee Membership					
Name	Age	Year Elected	Other Public Boards	Audit	Compensation & Human Capital	Finance	Governance & Corporate Responsibility	Risk	Technology
Greg Becker, CEO & President	55	2011	-						
Beverly Kay Matthews, Board Chair	64	2020	1	×			×	×	
Eric Benhamou	67	2005	2			×	C	×	
Elizabeth "Busy" Burr	61	2022	2	×					×
Richard Daniels	68	2021	1	×	×			×	C
Alison Davis	61	2021	2	×	×				×
Joel Friedman	75	2005	-			C	×		
Thomas King	62	2022	3		×				
Jeffrey Maggioncalda	54	2012	1		×				×
Mary Miller	67	2016	-	C		×		×	
Kate Mitchell	64	2010	1			×		C	×
Garen Staglin	78	2012	-		C		×	×	

Note: Committee membership as of March 3, 2023. C = chair.

Becker was appointed president and CEO in 2011. He joined the company in 1993 as part of the Northern California Technology Division and since then served in a number of executive and senior management roles, including division manager of venture capital, chief banking officer, chief operating officer, and president of the bank.

Benhamou was founder and general partner of Benhamou Global Ventures, LLC, an investment firm focused on innovative high-tech companies around the world, which he founded in 2003. He also sat on the boards of various public and private technology companies and various educational and philanthropic organizations. His former roles included CEO of 3Com Corp. and Palm, Inc.

Burr was the former president and chief commercial officer of Carrot, Inc., a business-to-business-to-consumer digital health company, where she built the go-to market strategy and team, including sales, marketing, communications, financial planning, account management, business development, and partnerships. In 2023, the Board of Rite Aid Corp. appointed Burr interim CEO while it conducted a search for a permanent CEO. She previously served in various executive management roles at Humana Inc., Citi Ventures, Morgan Stanley, Credit Suisse First Boston, and Gap, Inc.

Daniels retired from Kaiser Permanente, a nonprofit integrated managed care consortium, in 2020; there, he served as executive vice president and chief information officer. During his 12 years at Kaiser Permanente, he held a number of leadership positions, including senior vice president for both enterprise shared services and business information. Other previous roles include a variety of management positions at Capital One Financial Corp., JPMorgan Chase & Co., NetServ, Inc., and DataPoint Corp..

Davis was managing partner and cofounder of Blockchain Coinvestors, a blockchain venture investor and fund of funds. Davis had held a variety of management positions in the finance and venture capital (VC) industries, including at Belvedere Capital Partners, Inc., Barclays Global Investors, A.T. Kearney, and McKinsey & Company.

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Exhibit 16 (continued)

SVB Financial Holdings Board of Directors, 2022

Friedman retired in 2005 from Accenture PLC, a public company global management-consulting firm, where he held the position of president of the business process outsourcing organization. Over the course of a 34-year career with Accenture, Friedman held a variety of senior leadership roles, including managing partner, banking and capital markets; managing general partner, Accenture Technology Ventures; and founder, Accenture strategy consulting practice. He also served as director on various companies and organizations, including Neustar, Inc.; FTV Capital; Endeca Technologies; EXL Services (advisory director); Junior Achievement of Northern California; Accenture; Seisent, Inc.; and Calico Commerce, Inc.

King was the former CEO of investment banking at Barclays PLC, where he oversaw the banking and markets businesses and also served as the chair of the investment banking executive committee and a member of the Barclays Group executive committee. King previously held several senior positions at Citigroup, including global head of mergers and acquisitions, head of investment banking for the EMEA region, and head of corporate and investment banking for the EMEA region. In 2023, he was an operating partner at Atlas Merchant Capital LLC, a global investment firm.

Maggioncalda was currently (in 2023) the CEO of Coursera, Inc., an online education company, and also served on Coursera's board of directors. He previously served as the founding CEO and director of Financial Engines, Inc., a public independent investment advisory firm. Subsequent to his tenure at Financial Engines, Maggioncalda served as a senior adviser to McKinsey & Company, a global management-consulting firm.

Matthews retired from Ernst & Young LLP (E&Y), a multinational professional services firm, in 2019; there, she served as vice chair of the Americas executive board, a member of the global practice group, and the managing partner of the West region. During 36 years at E&Y, she held a variety of leadership positions, including senior advisory partner for key clients in the technology and transportation sectors.

Miller was the former under secretary for domestic finance for the US Department of the Treasury (US Treasury), a position that she held, following her confirmation by the US Senate, from March 2012 to September 2014. Miller also served as assistant secretary of the Treasury for financial markets from February 2010 to March 2012. Prior to joining the US Treasury, she led T. Rowe Price Group's fixed income division; T. Rowe Price was her employer for 26 years.

Mitchell was a partner and cofounder of Scale Venture Partners, a VC firm that invested in enterprise software companies, and was instrumental in building the firm's team and strategic direction. She held various directorships and other senior management positions throughout her career. Prior to founding Scale, Mitchell spent 20 years in various leadership positions at Bank of America in technology, finance, and operations.

Staglin was the founder and proprietor of Staglin Family Vineyard, founded in 1985 in Napa Valley. Over the past 40 years, Staglin had also held a variety of positions in the financial services and insurance industries, as well as directorships of various public and private companies.

Note: EMEA = Europe, the Middle East, and Africa.

Source: Adapted from SVB Financial Group SEC Schedule 14A, 2023 Proxy Statement, March 3, 2023, https://d18rn0p25nwr6d.cloudfront.net/CIK-0000719739/7d992e5b-9272-4bc8-a5cf-5a8c6b8c5251.pdf (accessed June 12, 2023).

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Appendix

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A defining function of banks was maturity transformation (also known as maturity mismatch): borrowing short term and lending long term.[1] In other words, banks invested in long-term assets (e.g., loans to clients and securities, like US notes and bonds), funded by short-term liabilities (largely deposits). "In textbook models, banks engage in maturity transformation to earn the average difference between the long- and short-term rates—the term premium—but this exposes them to interest rate risk," wrote scholars. "An unexpected increase in the short rate makes banks' interest expenses rise relative to interest income, pushing down net interest margins and depleting banks' capital."[2] Another scholar added, "When short-term rates increase, banks' cost of funding rises, and with fixed-rate assets, their profit margins shrink, which drags down their stock prices."[3] Wrote a senior executive and a visiting scholar at the St. Louis Federal Reserve,

> Borrowing over short periods and lending for long periods generally allows banks to make money, because long-term interest rates are usually higher than short-term interest rates. But this strategy carries the risk that interest rates will rise, reducing the value of a bank's long-term fixed-rate assets—usually loans or bonds—because higher interest rates reduce the present value of the payoffs to those loans or bonds. Potential decline in asset values because of an increase in interest rates is known as interest rate risk or duration risk, and it can reduce a bank's net worth (assets minus liabilities).[4]

Per one academic study, the average mismatch was about four years and historically stable; aggregate bank assets had an average estimated duration of 4.2 years, versus 0.4 years for liabilities.[5] Changes in a bank's liquidity and in interest rates could enlarge the mismatch between assets and liabilities.[6] Banks used asset-liability management (ALM) to manage their mismatch risk. ALM was

> the process by which banks price assets and liabilities…to maximize net interest income (NII) within the institution's risk parameters. In the normal course of business, banks book assets (primarily loans and securities) and fund [them] with liabilities (deposits and borrowings). These assets and liabilities have differing maturity dates, cash flows, and overall structures, and natural mismatches in structure occur. Banks use various measurement methodologies to determine if earnings or capital will be impacted by movements in interest rates (interest rate risk) or changes in market prices and conditions (liquidity risk).[7]

[1] Itamar Drechsler, Alexi Savov, and Philipp Schnabl, "Banking on Deposits: Maturity Transformation without Interest Rate Risk" (NBER Working Paper no. 24582, National Bureau of Economic Research, May 2018, revised January 2020, JEL No. E43,E52,G21,G31), https://www.nber.org/system/files/working_papers/w24582/w24582.pdf (accessed June 30, 2023).

[2] https://www.nber.org/system/files/working_papers/w24582/w24582.pdf.

[3] Pascal Paul, "Banks, Maturity Transformation, and Monetary Policy" (working paper 2020-07, Federal Reserve Bank of San Francisco, 2022), https://doi.org/10.24148/wp2020-07 (accessed June 30, 2023).

[4] Christopher J. Neely and Michelle Clark Neely, "Interest Rate Risk, Bank Runs and Silicon Valley Bank," St. Louis Federal Reserve, May 11, 2023, https://www.stlouisfed.org/publications/regional-economist/2023/may/interest-rate-risk-bank-runs (accessed June 30, 2023).

[5] https://www.nber.org/system/files/working_papers/w24582/w24582.pdf.

[6] Kevin Schalk, Ivan Cilik, and Sean Statz, "It's Time to Review Your Financial Institution's Liquidity and ALM Practices: Leverage Your Data to Mitigate Risk," Baker Tilly, March 30, 2023, https://www.bakertilly.com/insights/leverage-your-data-to-mitigate-risk (accessed June 29, 2023).

[7] "ALM Basics: What Is Asset Liability Management?," *Banc Path Asset Management Group, Inc.* (blog), https://www.ccbfinancial.com/publications/amg/alm-basics/alm-basics-what-is-asset-liability-management-september-2011 (accessed June 29, 2023).

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